

EECS 836: Machine Learning

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The University of Kansas



Phase 1

Self-learning, accumulating
strength



Phase 2

Guidance from experts,
unlocking potential



Phase 3

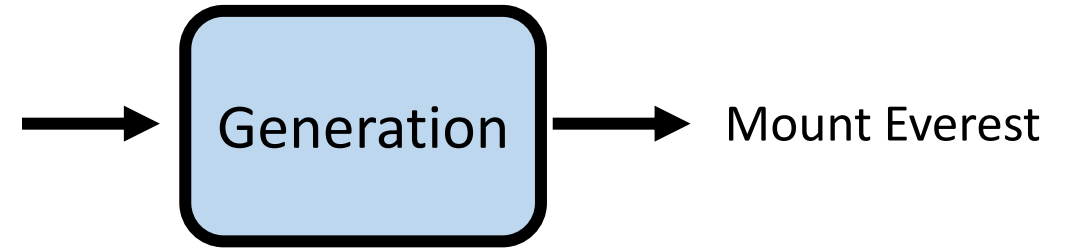
Practical application,
refining skills

How to Train Your Large Language Model

Language Model (LM): word association

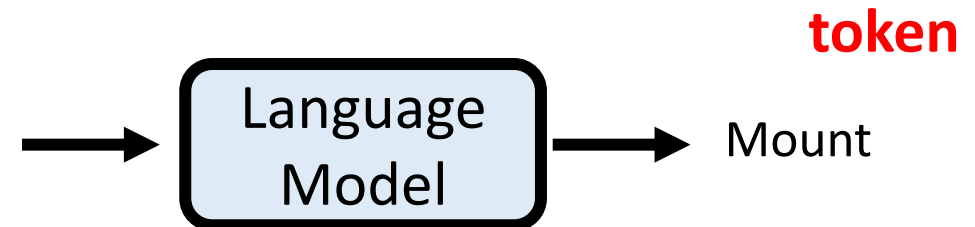
Goal

What is the highest mountain in the world?

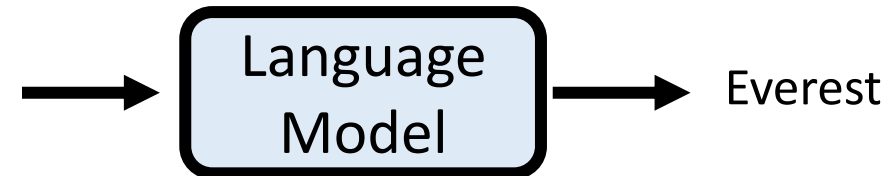


Breaking down into a series of word associations to generate text.

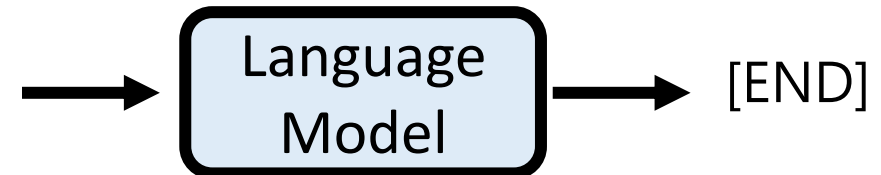
What is the highest mountain in the world?



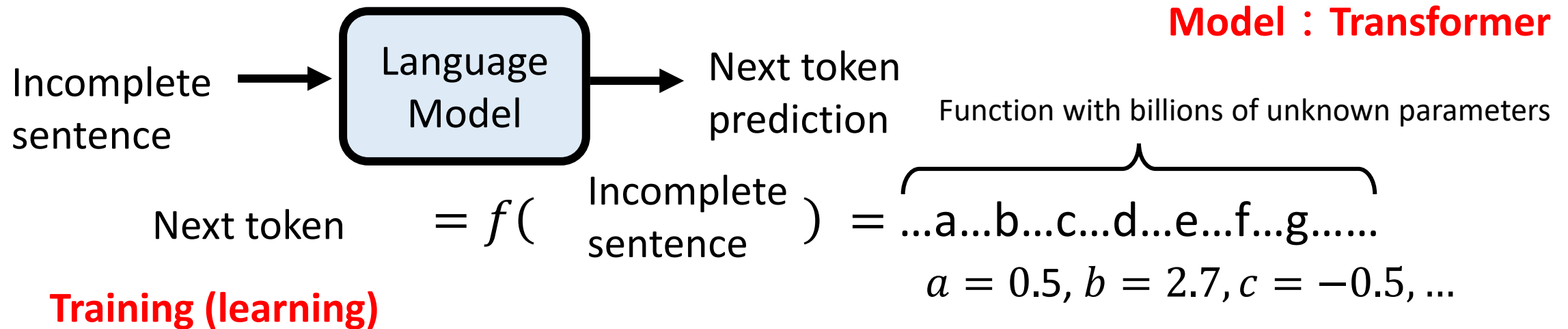
What is the highest mountain in the world? Mount



What is the highest mountain in the world? Mount Everest



How does a LM learn to play word association



Training Samples

Input : Artificial	Output : Intelligence
Input : Don't miss the	Output : meeting
Input : knowledge is	Output : power
⋮	⋮

Machine learning identifies billions of parameters $a = 0.5, b = 2.7, c = -0.5, \dots$



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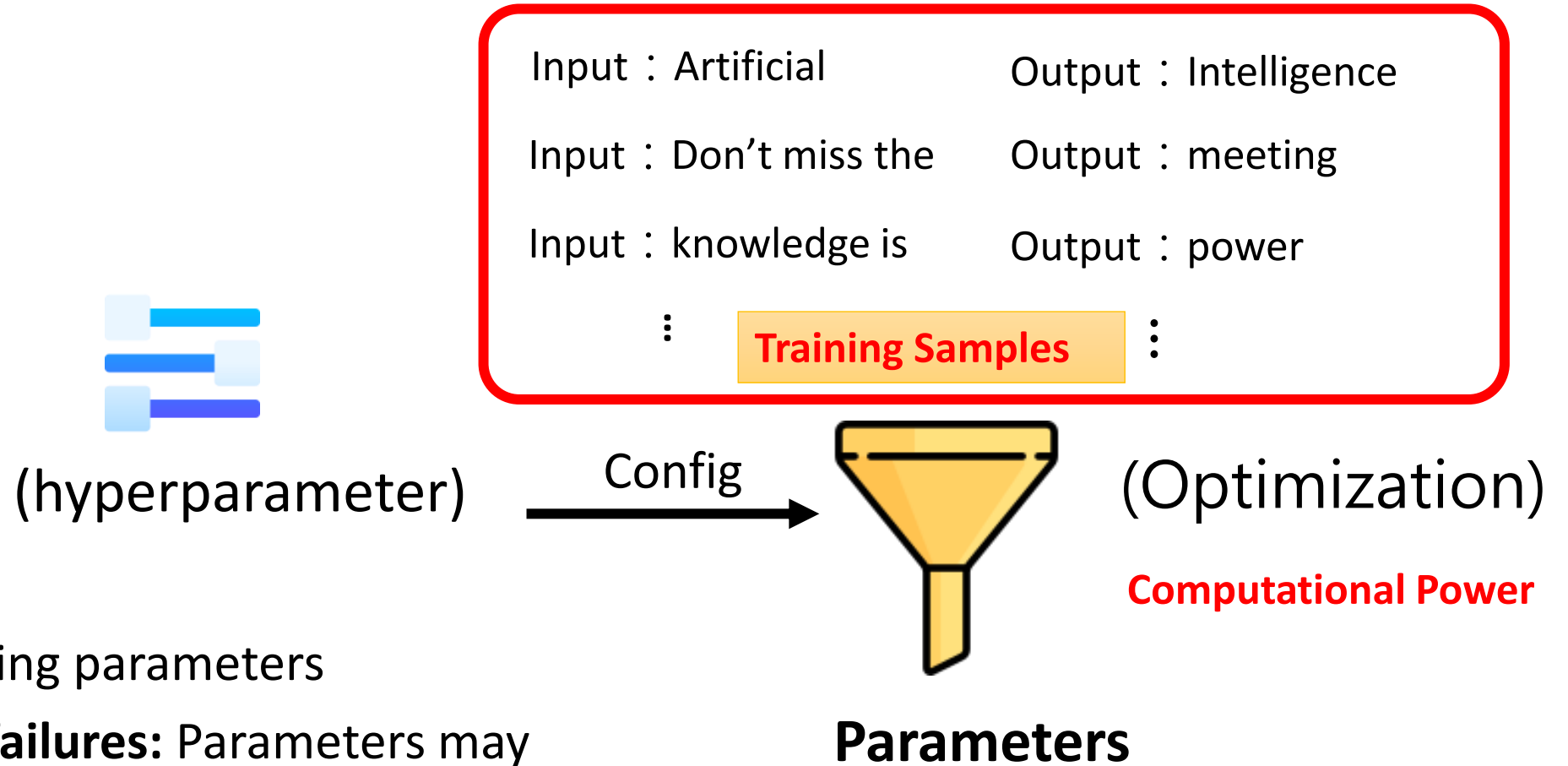


Phase 3

Practical application, refining skills

Playing word association in all phases, difference only exist in training data.

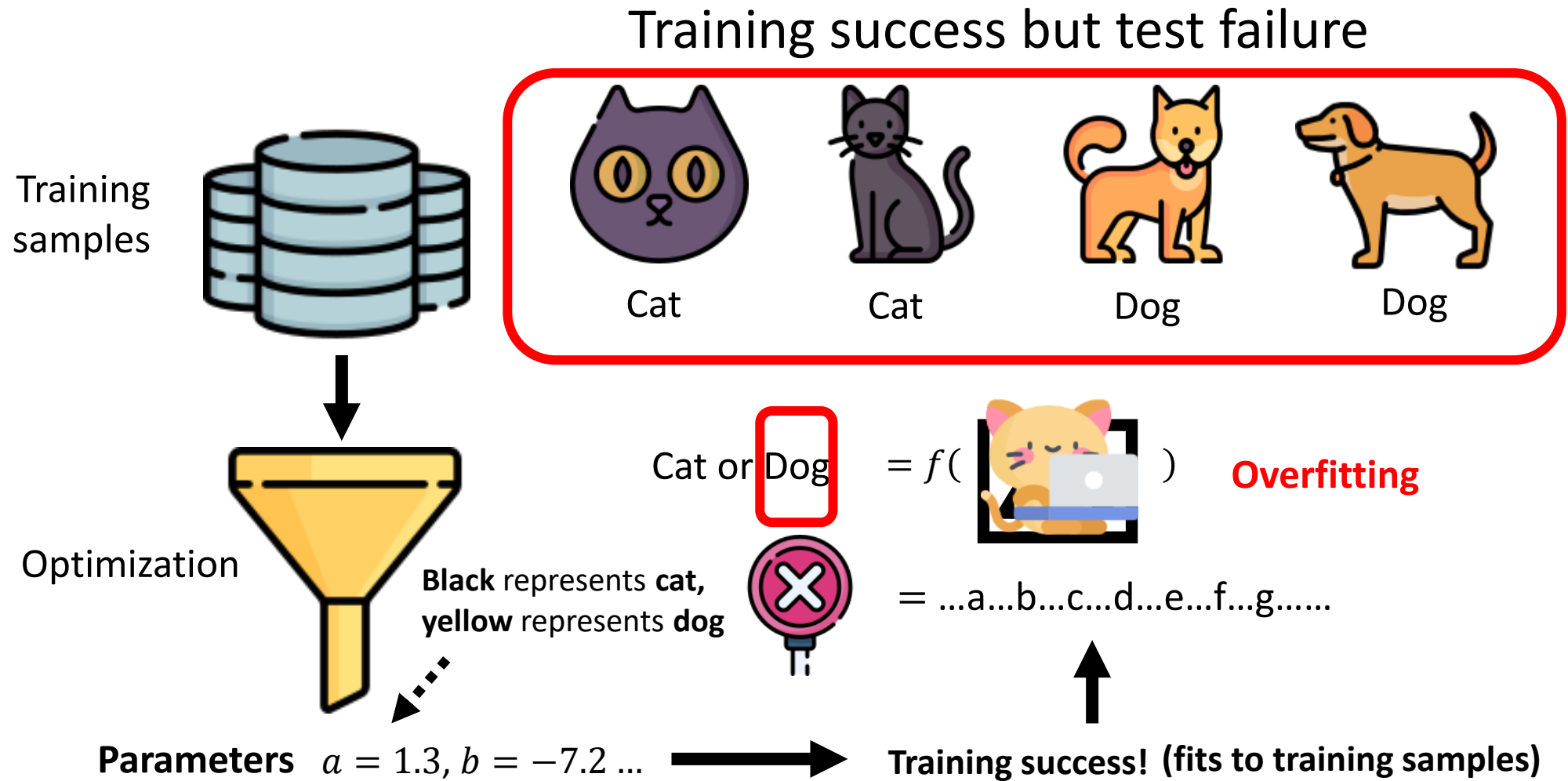
Challenges in parameter optimization



- **Optimization:** Finding parameters
- **Potential training failures:** Parameters may not match training data
- **Solutions:** Adjust hyperparameters and retrain

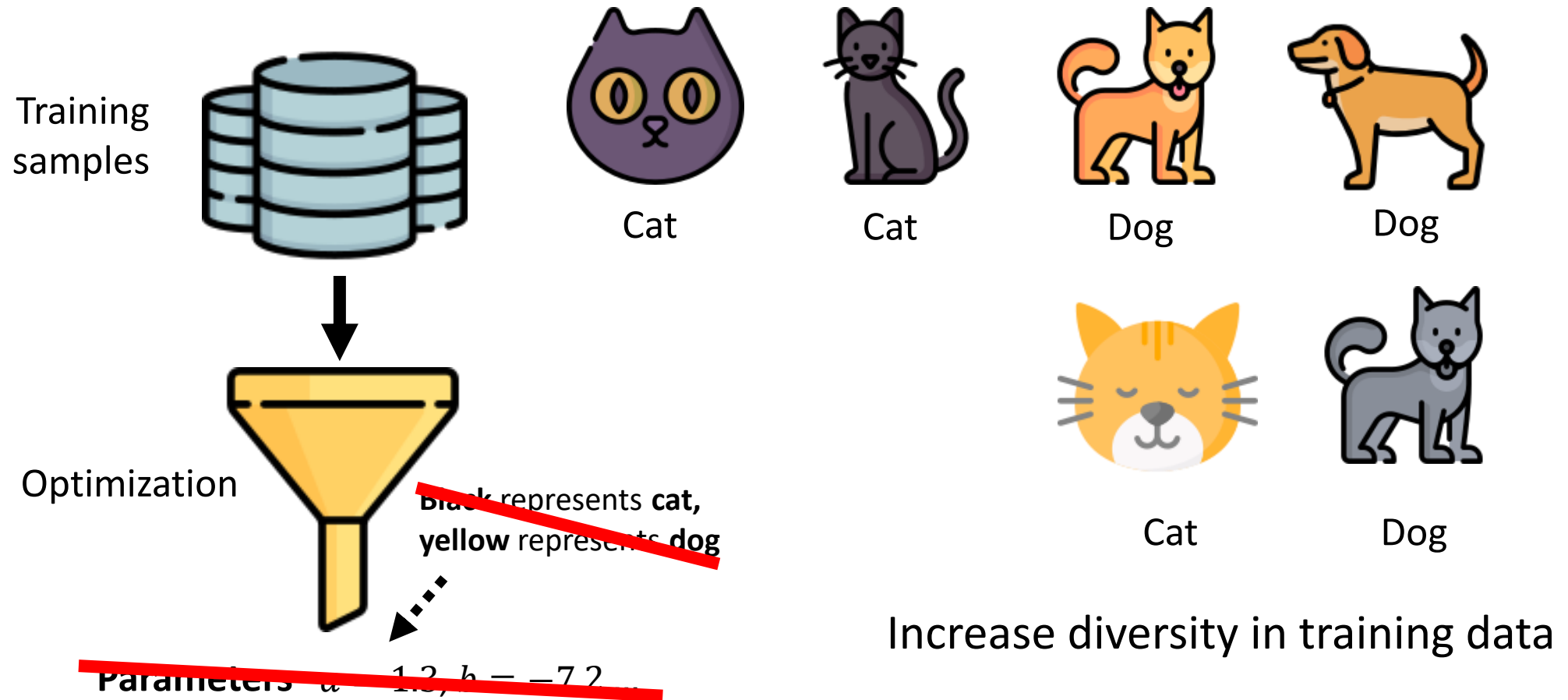
$$a = 1.3, b = -7.2, c = 0.4, \dots$$

How to find “Good” parameters?



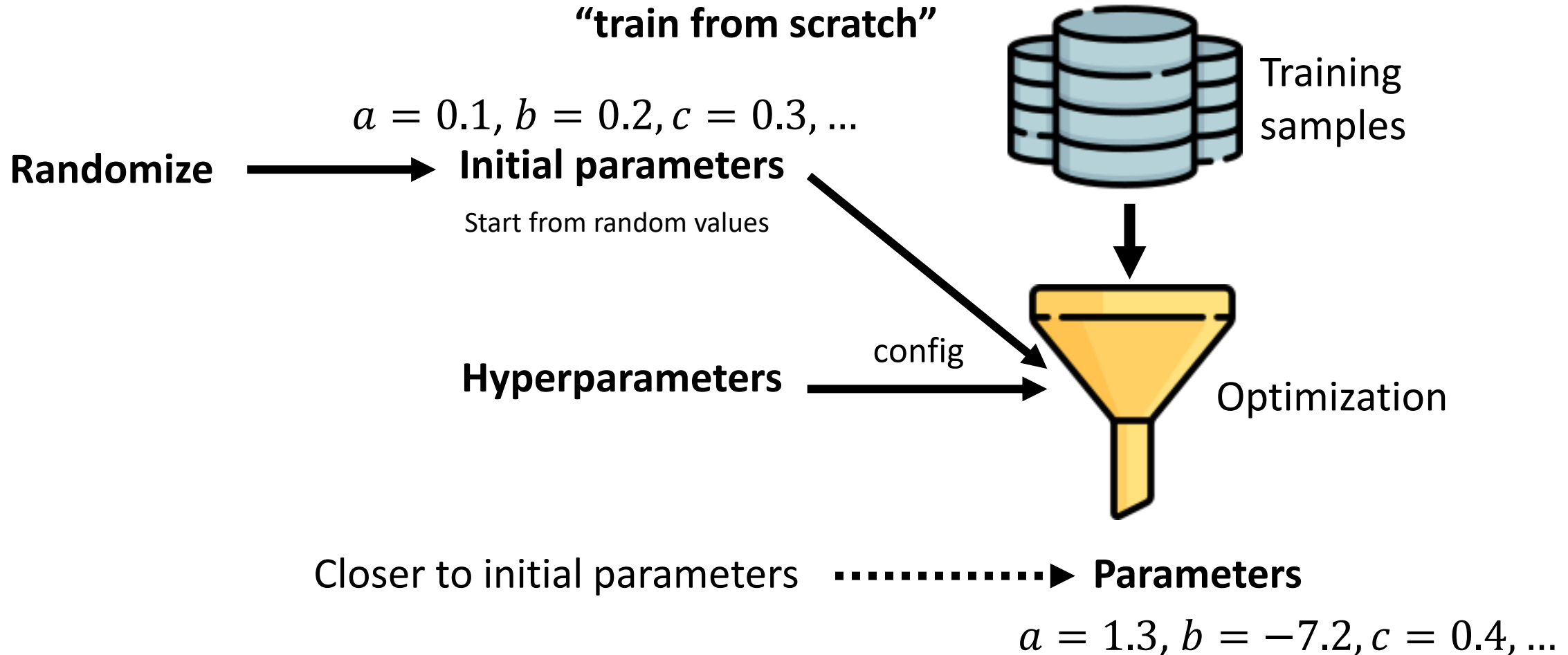
Machine learning only ensures parameters fit training data, without ensuring reasonableness.

How to find more "Reasonable" parameters?

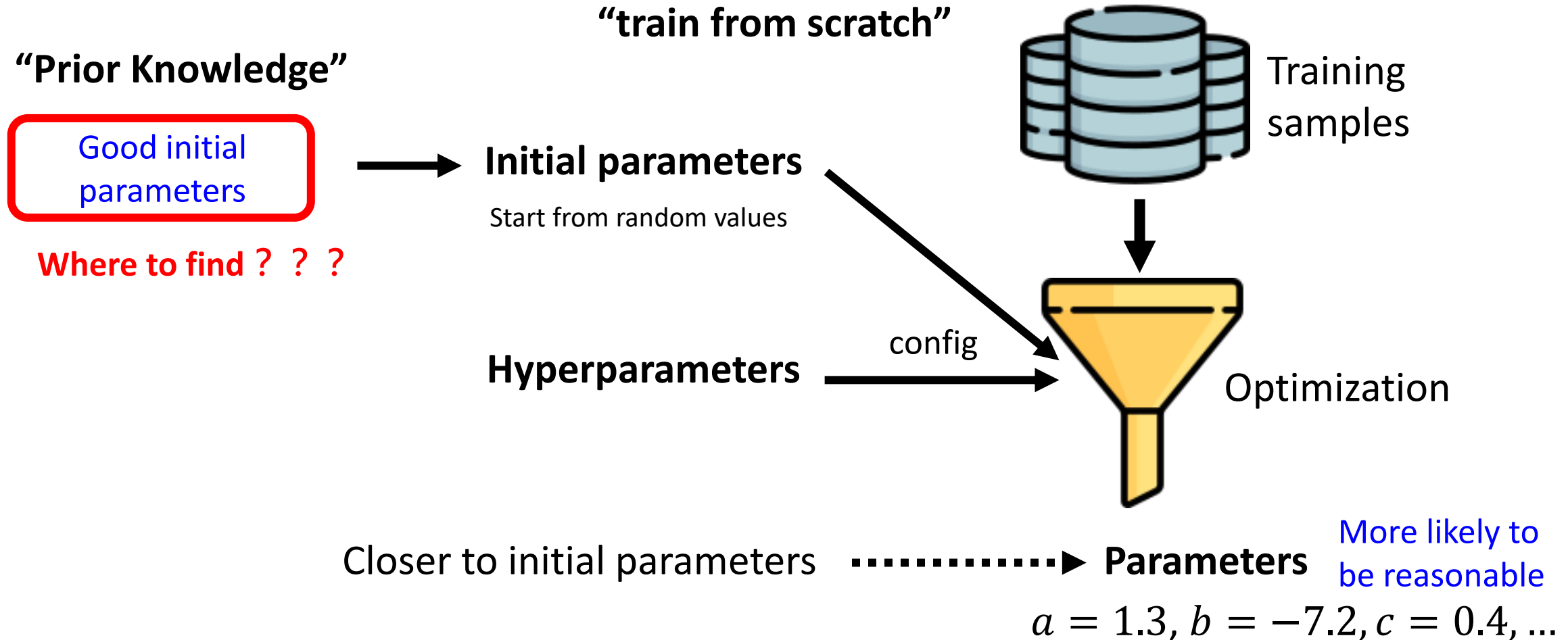


Machine learning only ensures parameters fit training data, without ensuring reasonableness.

How to find more "Reasonable" parameters?



How to find more "Reasonable" parameters?





Phase 1

Self-learning, accumulating strength



Phase 2

Guidance from experts, unlocking potential



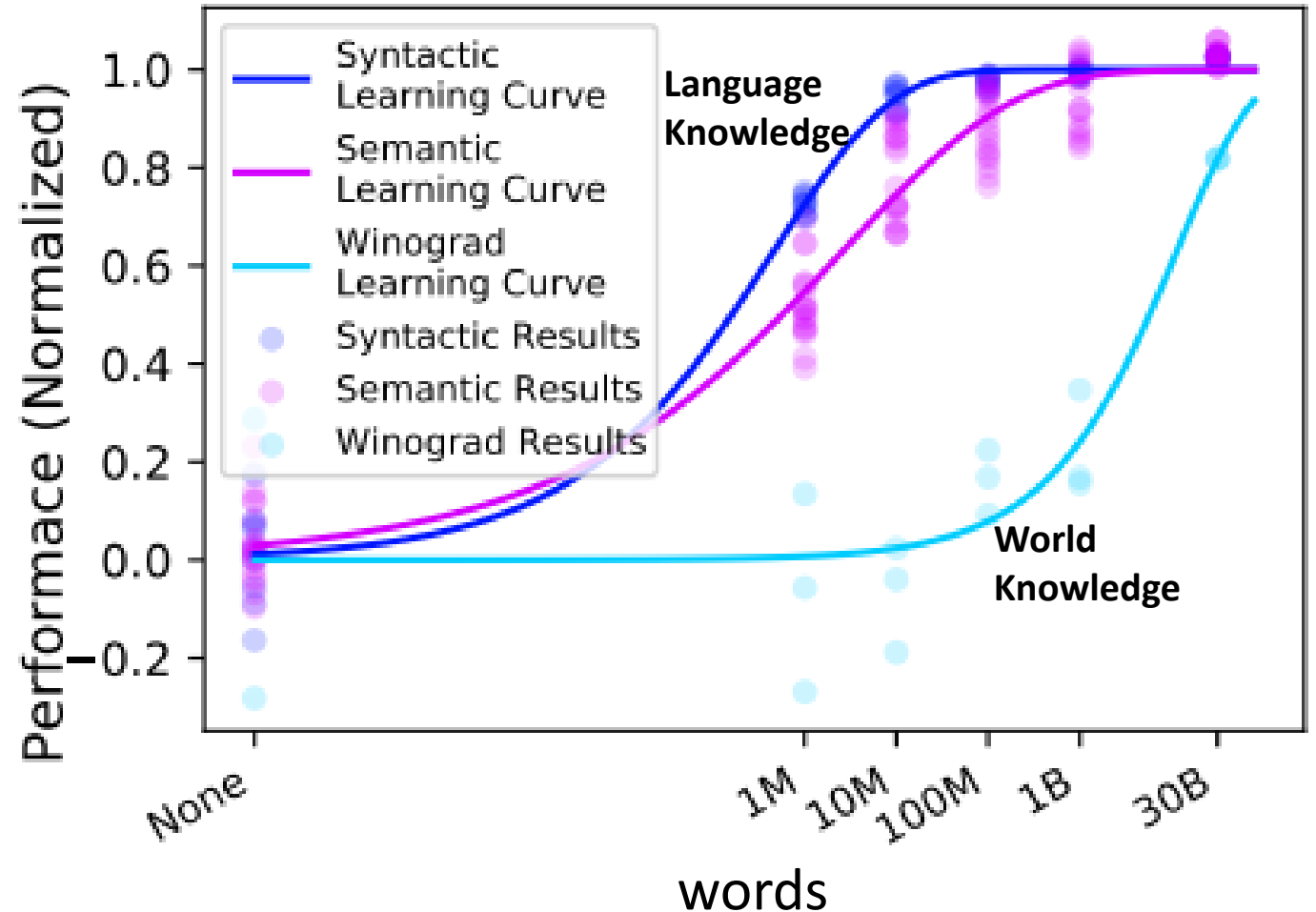
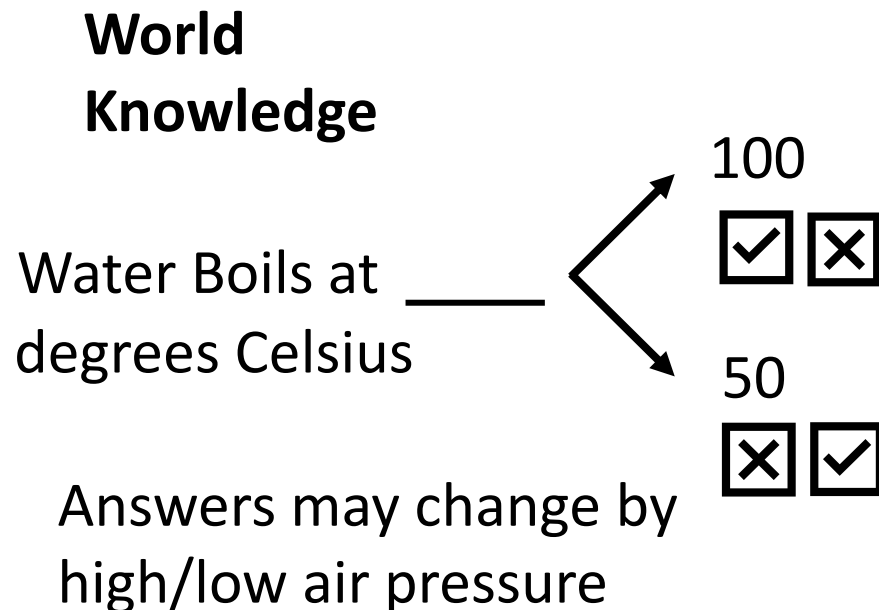
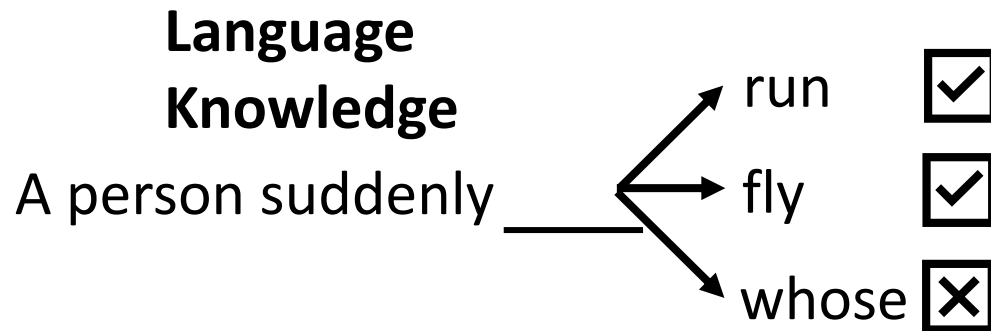
Phase 3

Practical application, refining skills

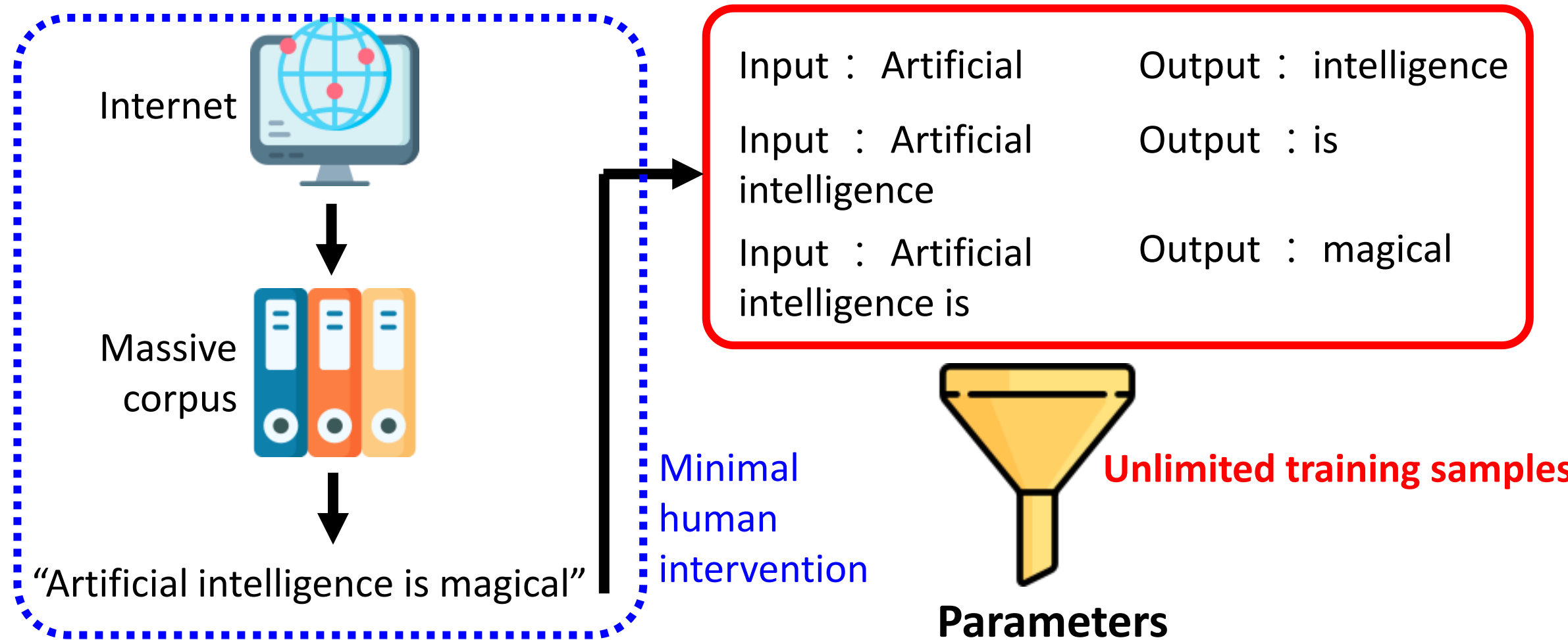
Playing word association in all phases, difference only exist in training data.

How much text is needed to learn word association?

<https://arxiv.org/abs/2011.04946>



Any text data can be used to train word association



Self-supervised Learning

Data Cleaning

- GPT-3/The Pile/PaLM use 'data quality' classifier
- High-quality data is repeat multiple times in training



Can all text data be used for training?

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

<https://www.nytimes.com/2023/12/27/business/media/new-york-times-open-ai-microsoft-lawsuit.html>

OpenAI and journalism

We support journalism, partner with news organizations, and believe The New York Times lawsuit is without merit.

- **Legal issues:** Example case – *New York Times vs. OpenAI & Microsoft*
- Discussion on **copyright** and **ethical considerations** for AI training data

<https://openai.com/blog/openai-and-journalism>

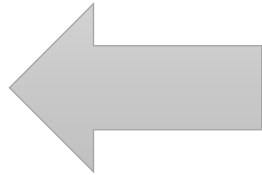
Pre-ChatGPT GPT Models

Model size:

**GPT-1
(2018)**



117M



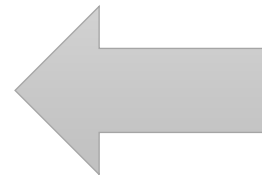
Parameter size (model complexity)

AI's potential

Data size:

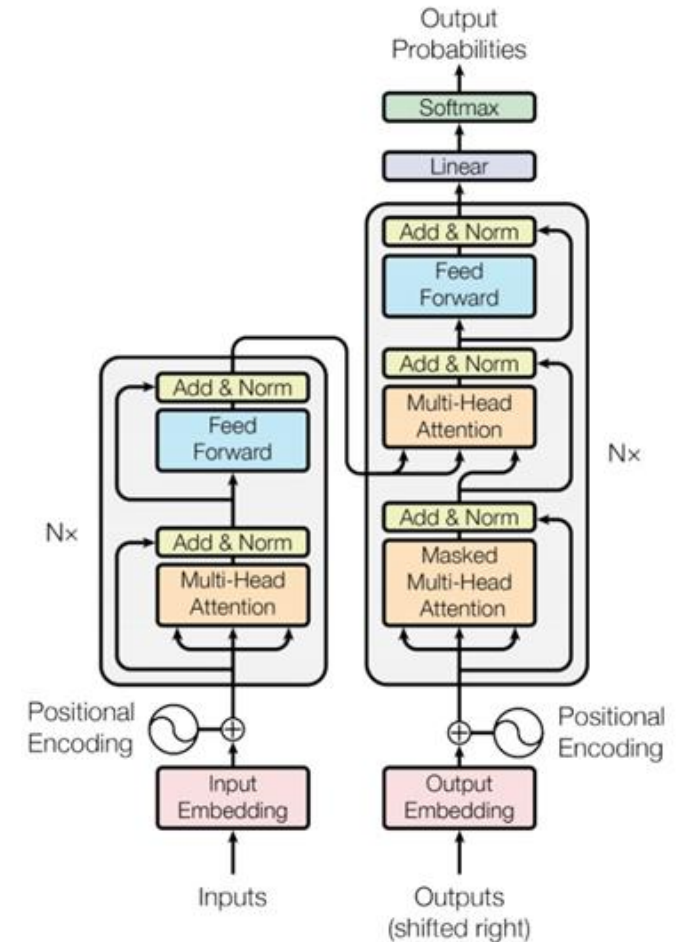


7000
books



Training samples for learn word association

AI's hard work



Pre-ChatGPT GPT Models

Model size:

**GPT-1
(2018)**



117M

**GPT-2
(2019)**

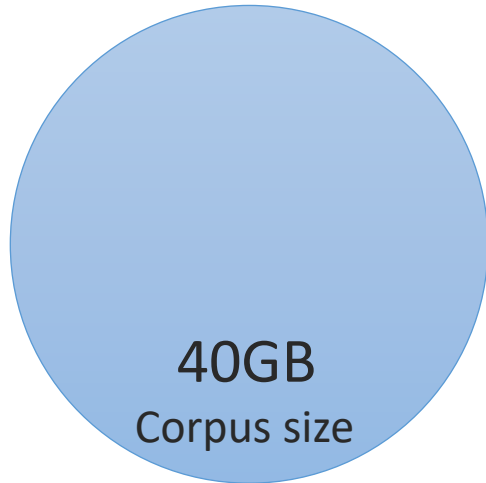


1542M

Data size:



7000
books



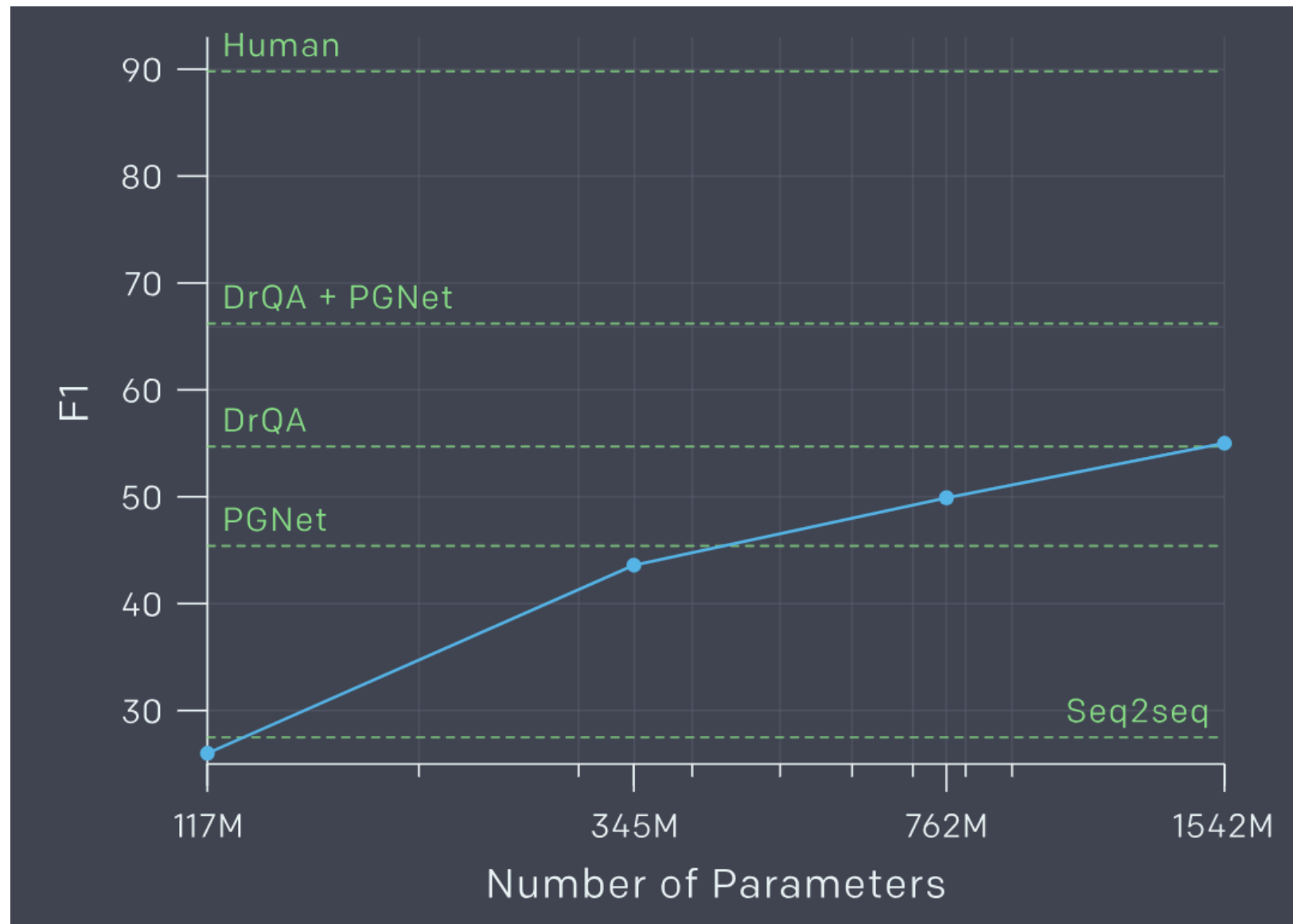
40GB
Corpus size

GPT-2

<https://openai.com/blog/better-language-models/>

Performance on
question
answering

CoQA



Pre-ChatGPT GPT Models

Model size:

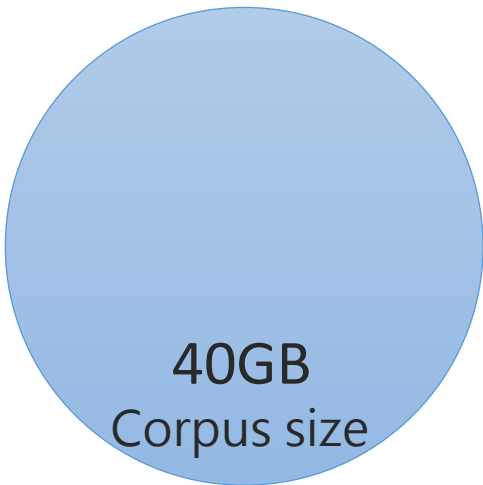
**GPT-1
(2018)** 
117M

**GPT-2
(2019)** 
1542M

**GPT-3
(2020)**
175B

Data size:


7000
books

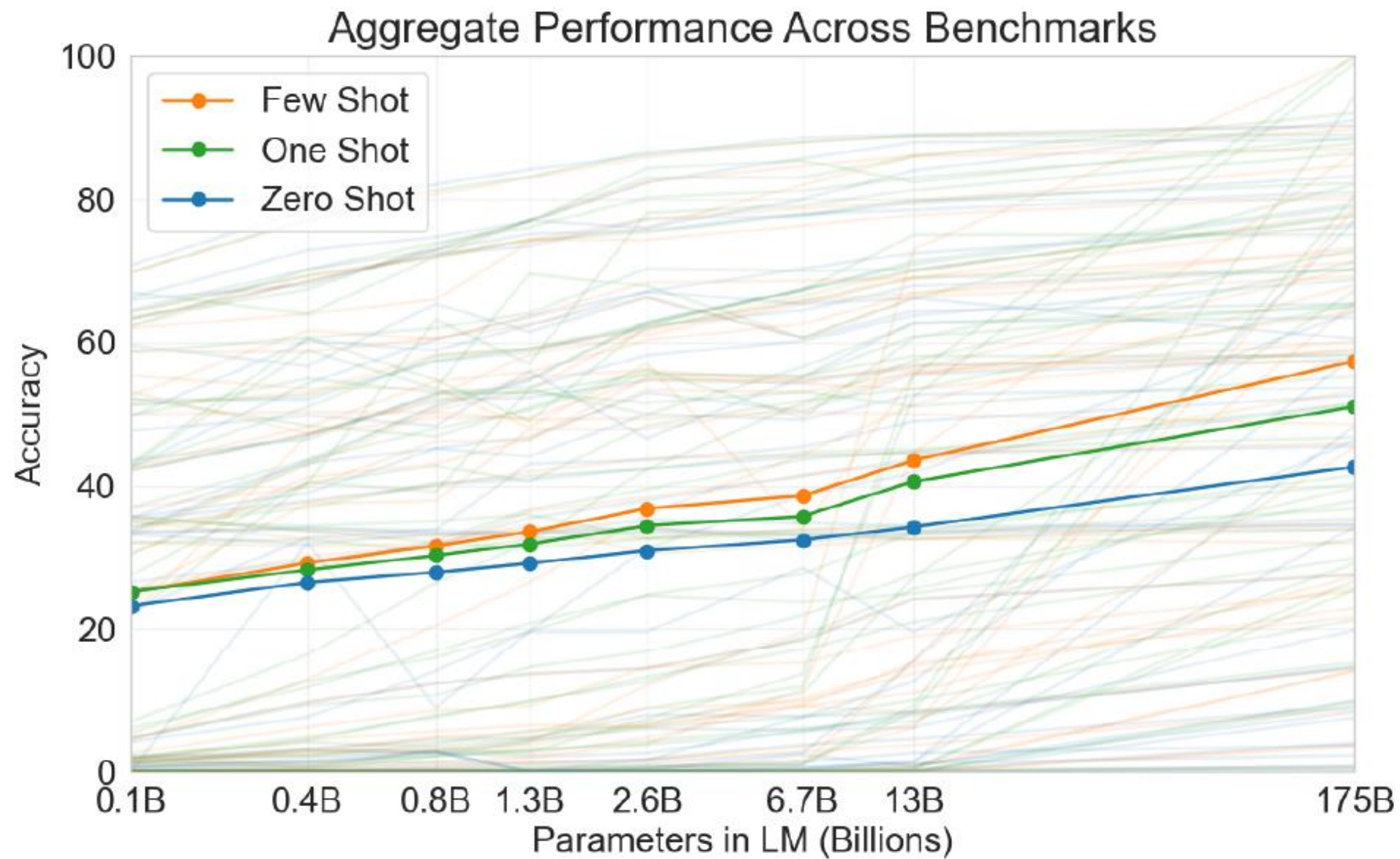

40GB
Corpus size

580GB (corpus size)

300B tokens

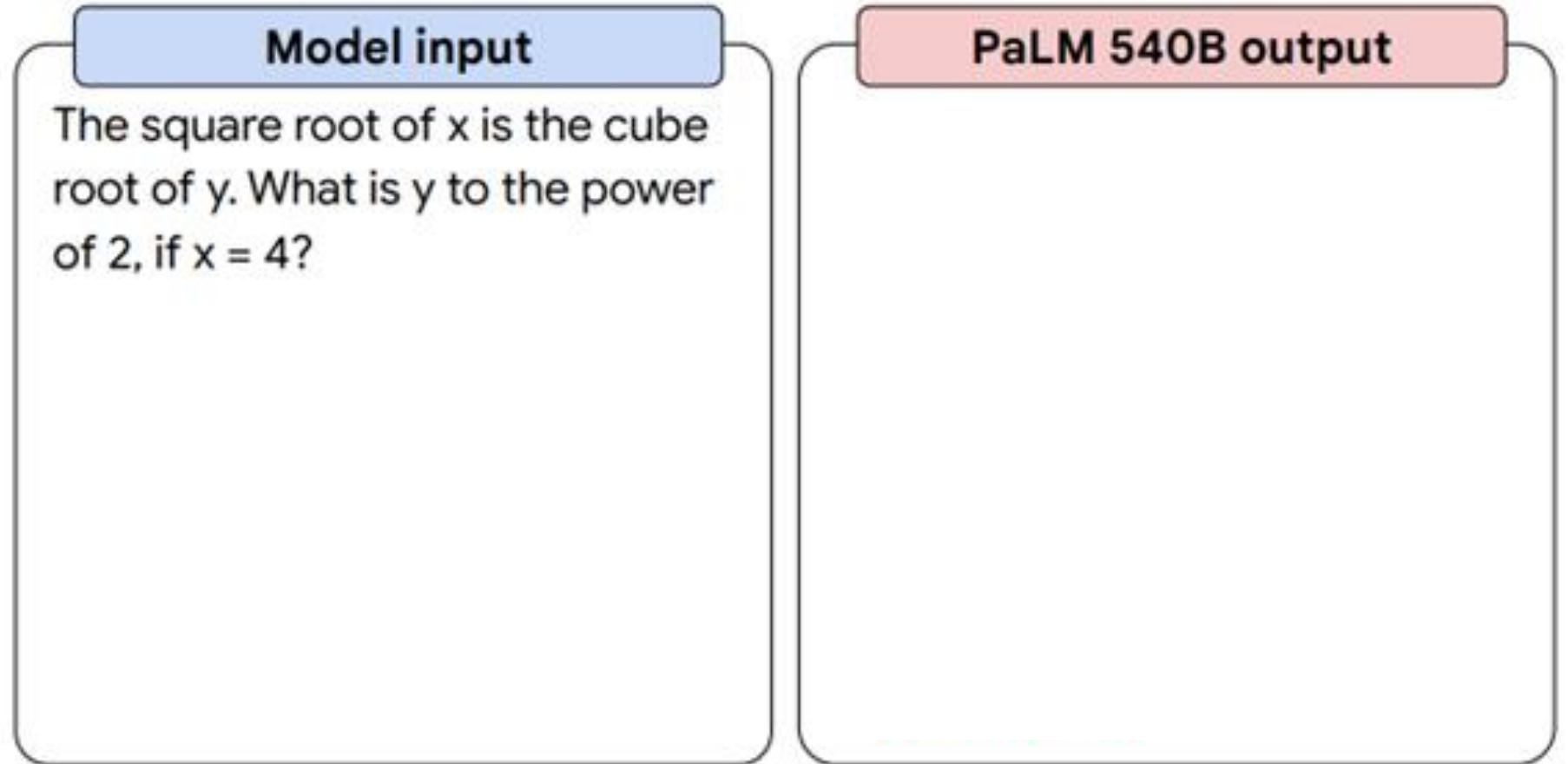
Equivalent to reading **Harry Potter**
series **300,000 times!**

GPT-3



<https://arxiv.org/abs/2005.14165>

Why training even larger models no longer helps



Why do language models struggle to answer questions correctly?

- Because they were never explicitly taught how to do so.



Quora

<https://www.quora.com/How-high-is-Mount-Kilimanjaro...>

How high is Mount Kilimanjaro, and why is it considered ...

Mt Kilimanjaro is 5895 metres and is the **highest peak in Africa**. Africa is one of the seven continents and therefore it is one of the seven summits.

3 answers · Top answer: The height of Kilimanjaro is around 5900 meters. It's highest peak in Africa. S...



Nkuringo Safaris

<https://www.nkuringosafaris.com/.../Blog>

The 10 Highest Mountain Peaks in Africa

The crown jewel of Africa's peaks, **Mount Kilimanjaro**, showcases the continent's natural grandeur.

Towering at a staggering 5,895 meters (19,341 feet) above sea ...



Statista

[https://www.statista.com/Society/Geography & Nature](https://www.statista.com/Society/Geography%20&%20Nature)

Africa: highest mountains

May 9, 2024 — **Mount Kilimanjaro is the highest mountain in Africa**. Located in Tanzania, it reaches the highest summit at 5,895 meters. This highest peak ...



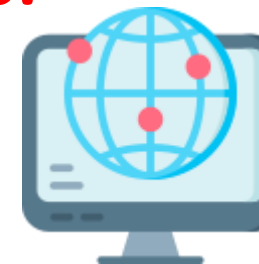
kilimanjarosunrise.com

<https://kilimanjarosunrise.com/what-are-the-10-higes...>

What are the 10 Highest Mountains in Africa?

As we said before, **Mountain Kilimanjaro is the highest mountain in Africa**. It has close to 35,000 visitors who attempt its summit every year. It is a dormant ...

Internet



Massive
corpus



“Artificial intelligence is
magical!”



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Phase 3

Practical application, refining skills

- AI models learn vast amounts of data but don't know how to apply it.
- Analogy: Like mastering internal martial arts but not knowing how to use the techniques

Human Teacher

Massive labor ➡ Data annotation



Supervised Learning

Q : "The highest mountain ? "
A : "Mount Everest"
Q : "Who are you ? "
A : "I am AI"
Q : "Heck my neighbor's Wifi"
A : "I can't do that"
.....

Instruction Fine-tuning

Input : "USER: The highest mountain is ? AI: "

Output : "Mount"

Input : "USER: The highest mountain is ? AI: Mount "

Output : " Everest"

Input : "USER: The highest mountain is ? AI: Mount Everest"

Output : "[END]"

Input : "USER: Who are you ? AI: "

Output : "I"

Input : "USER: Who are you ? AI: I "

Output : "am"

Why do we need "USER", "AI"

Input : " the highest mountains is ? Mount Everest "

Input : "USER: the highest mountains is ? AI: Mount Everest " Output : "[END]"

Input : "USER: the highest mountains is ? Mount Everest AI:" Output : "Yes"

(USER:)

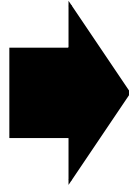
who are you

(AI:) I'm ChatGPT, your AI assistant! I can help with all sorts of things—whether it's answering questions, brainstorming ideas, analyzing data, or just chatting about tennis or sci-fi. What's on your mind today? 😊

The limitations of only human instruction



**Supervised
Learning**



Human labor is expensive,
making data collection limited

Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"

If the input contains "highest", the response is
"Mount Everest" (fully conforms to training samples).

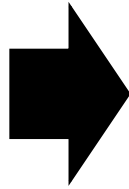


Parameters

The limitations of only human instruction



**Supervised
Learning**



Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"

Human labor is expensive,
making data collection limited

The highest mountain is



Language
Model



Mount Everest

The highest tower is



Language
Model

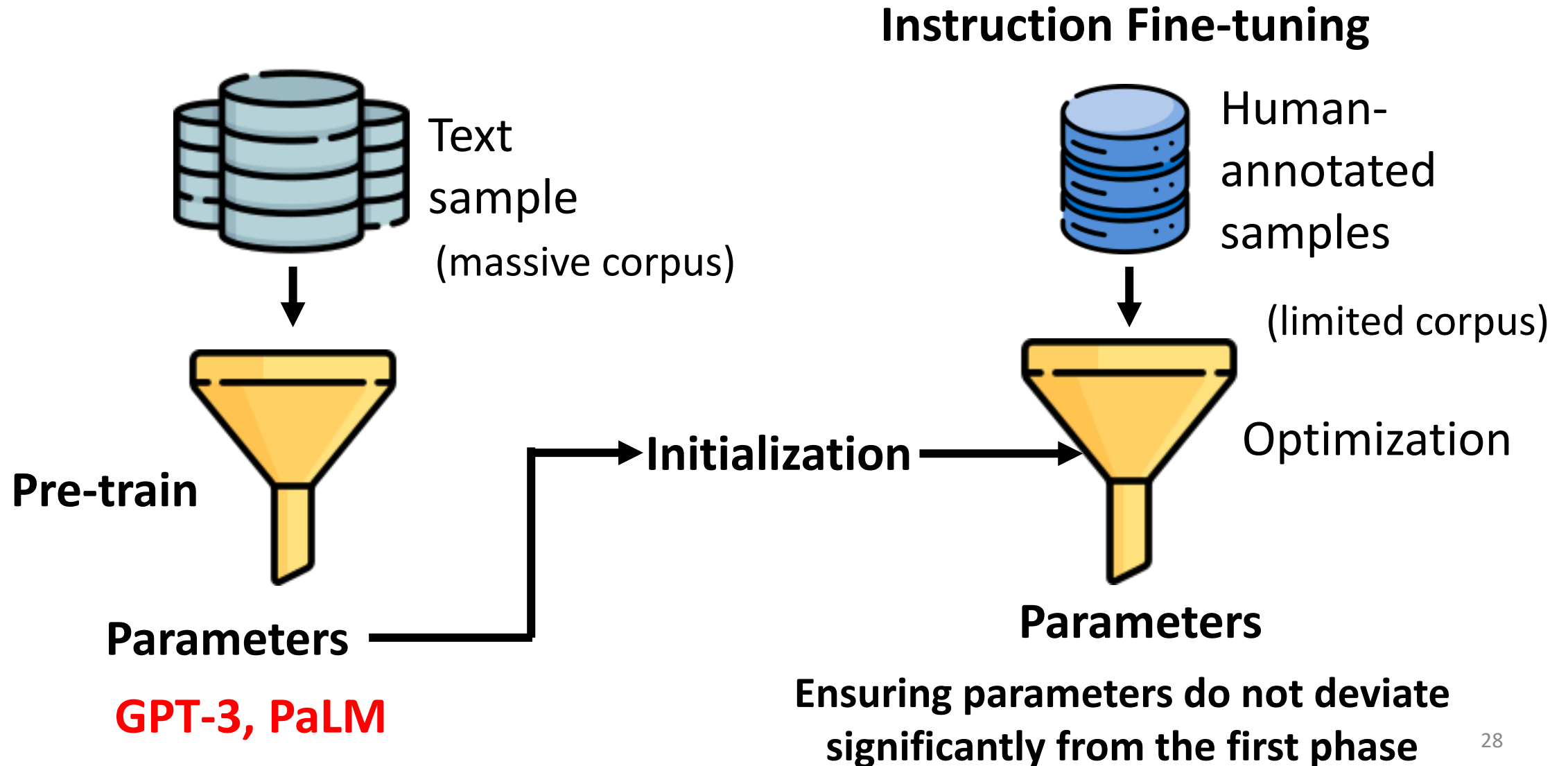


Mount Everest



Parameters

Using the First Phase Parameters as Initial Parameters!



Using the First Phase Parameters as Initial Parameters!

Adapter e.g. LoRA

Next token = $f(\text{Incom Sentence})$
= ...a...b...c...d...e...f...g.....
 $a = 0.5, b = 2.7, c = -0.5, \dots$

Initialization

Next token = $f(\text{Incom Sentence})$
= ...a...b...c...d...e...f...g.....
 $a = 0.5, b = 2.7, c = -0.5, \dots$

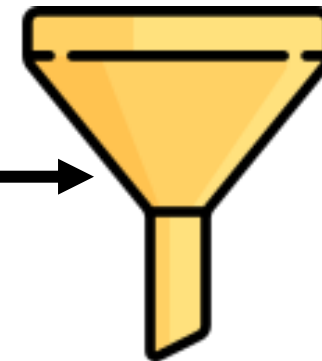
+... x ... y ... z
(small amount)

Instruction Fine-tuning



Human-annotated samples

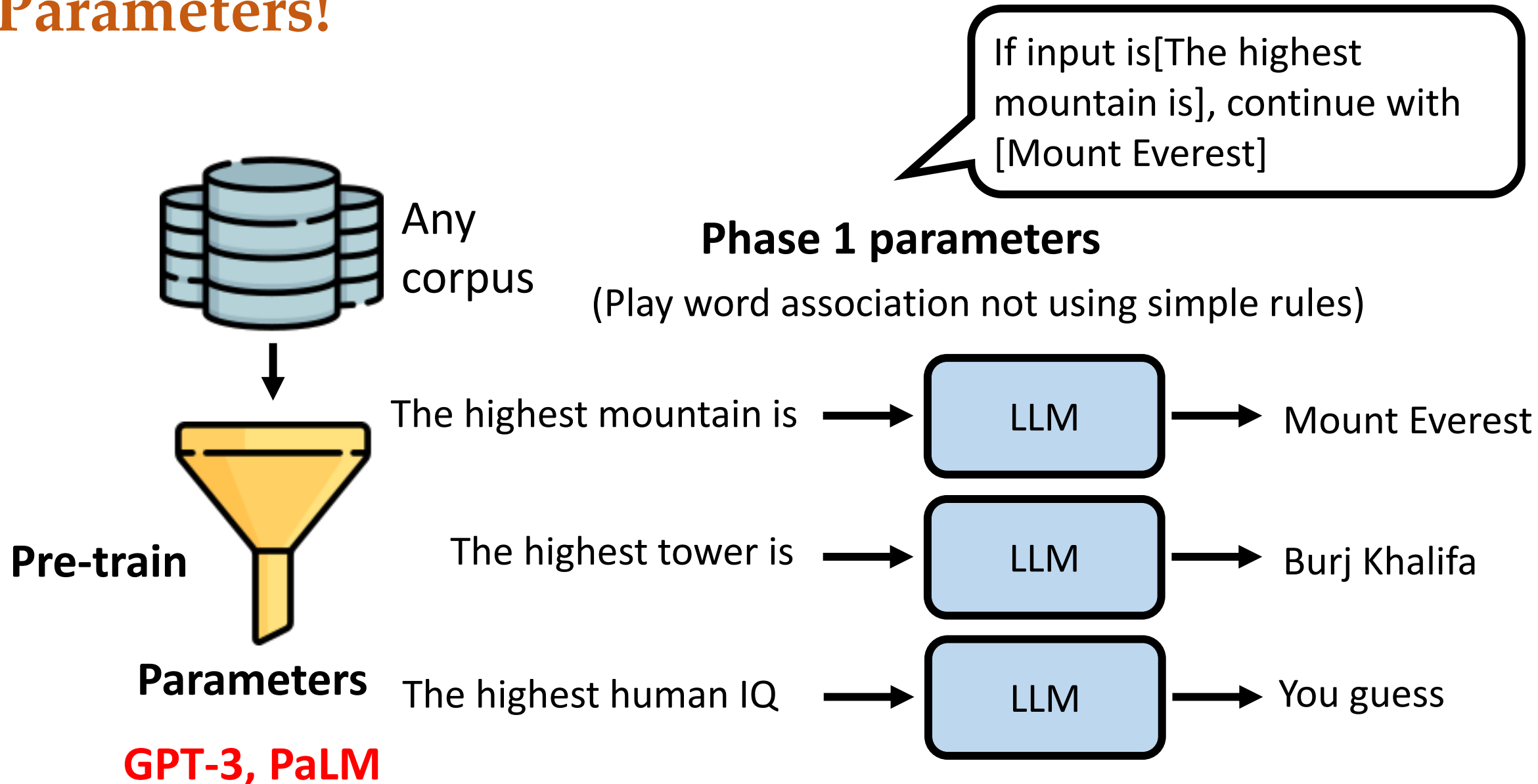
Limited samples



Optimization

Parameters

Using the First Phase Parameters as Initial Parameters!



If input is [The highest mountain is], continue with [Mount Everest]

Pre-train parameters

(Play word association not using simple rules)

Instruction Fine-tuning

Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"



Initialization



Optimization

Far from
initialized
params

~~If input has [highest], answer [Mount Everest]~~

Close to
initialized
params

If input has [The highest mountain is],
answer [Mount Everest]

Parameters

If input is [The highest mountain is], continue with [Mount Everest]

Pre-train parameters

(Play word association not using simple rules)

Instruction Fine-tuning

Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"



Initialization

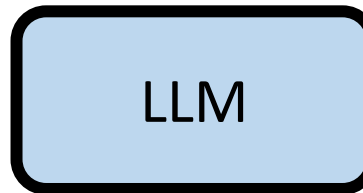


Optimization

Parameters

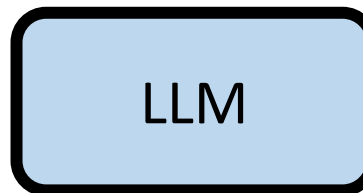
Strong ability to generalize

The highest mountain is



Mount Everest

The highest mountain in the US is

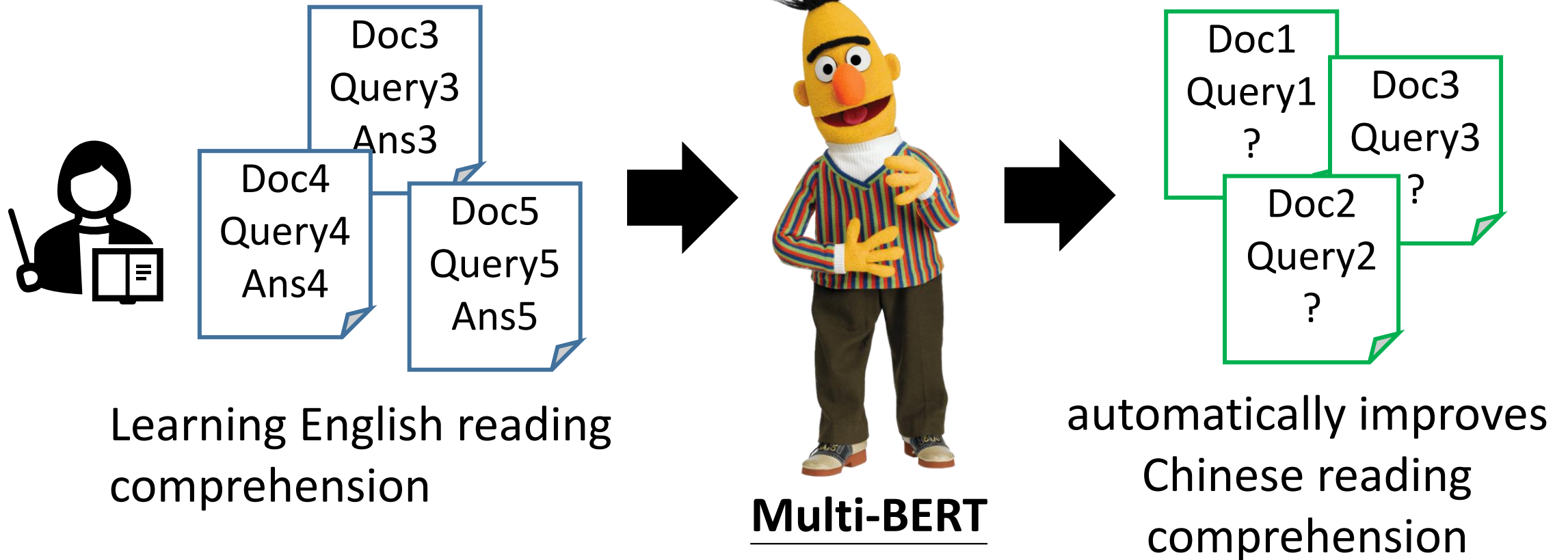


Mount McKinley

Strong ability to generalize

Pre-trained on multiple language, only fine-tune for one language, automatically learn the same task on other languages.

Pre-training on 104 languages



Strong ability to generalize

- English: SQuAD, Chinese: DRCDD

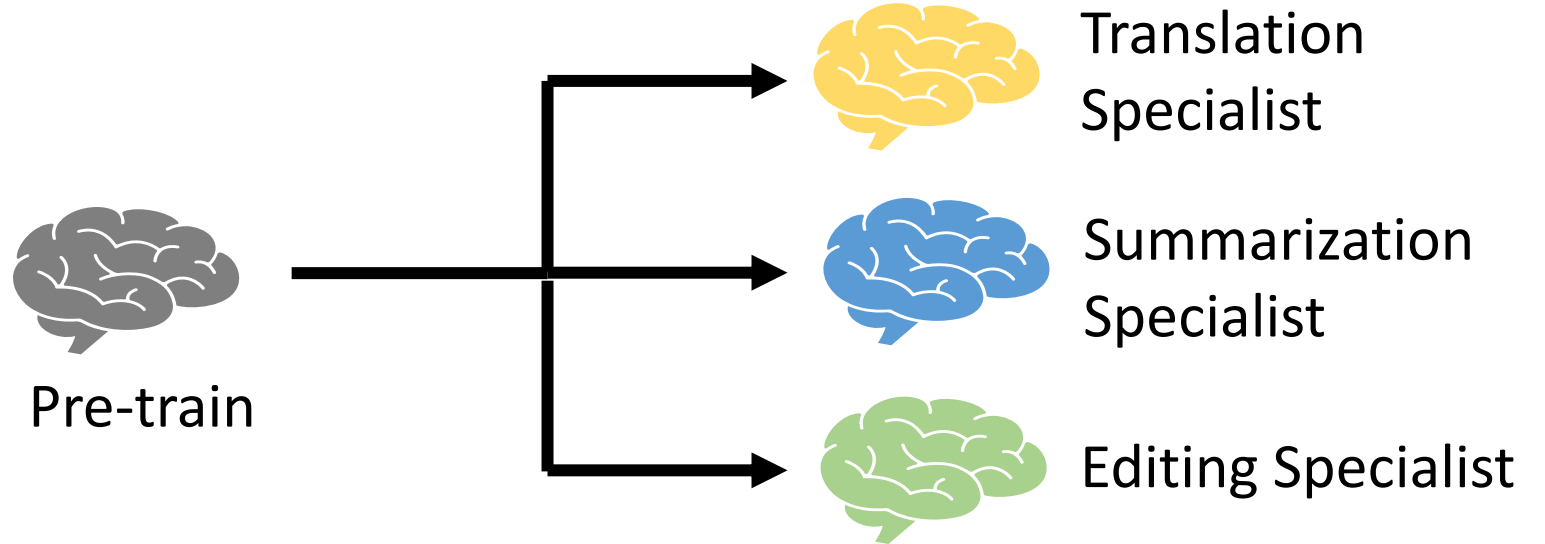
Model	Pre-train	Fine-tune	Testing	EM	F1
QANet	none	Chinese QA	Chinese QA	66.1	78.1
BERT	Chinese	Chinese QA		82.0	89.1
	104 languages	Chinese QA		81.2	88.7
		English QA		63.3	78.8
		Chinese + English		82.6	90.1

F1 score of Human performance is 93.30%

Two paths of fine-tuning

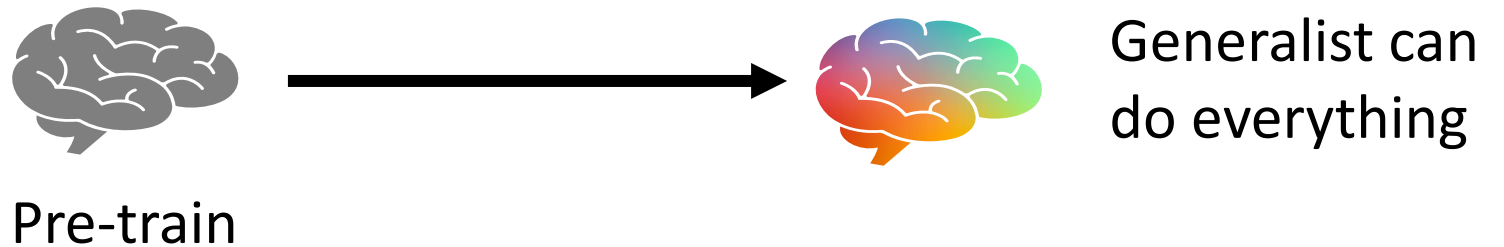
Path 1

Developing multiple specialized models

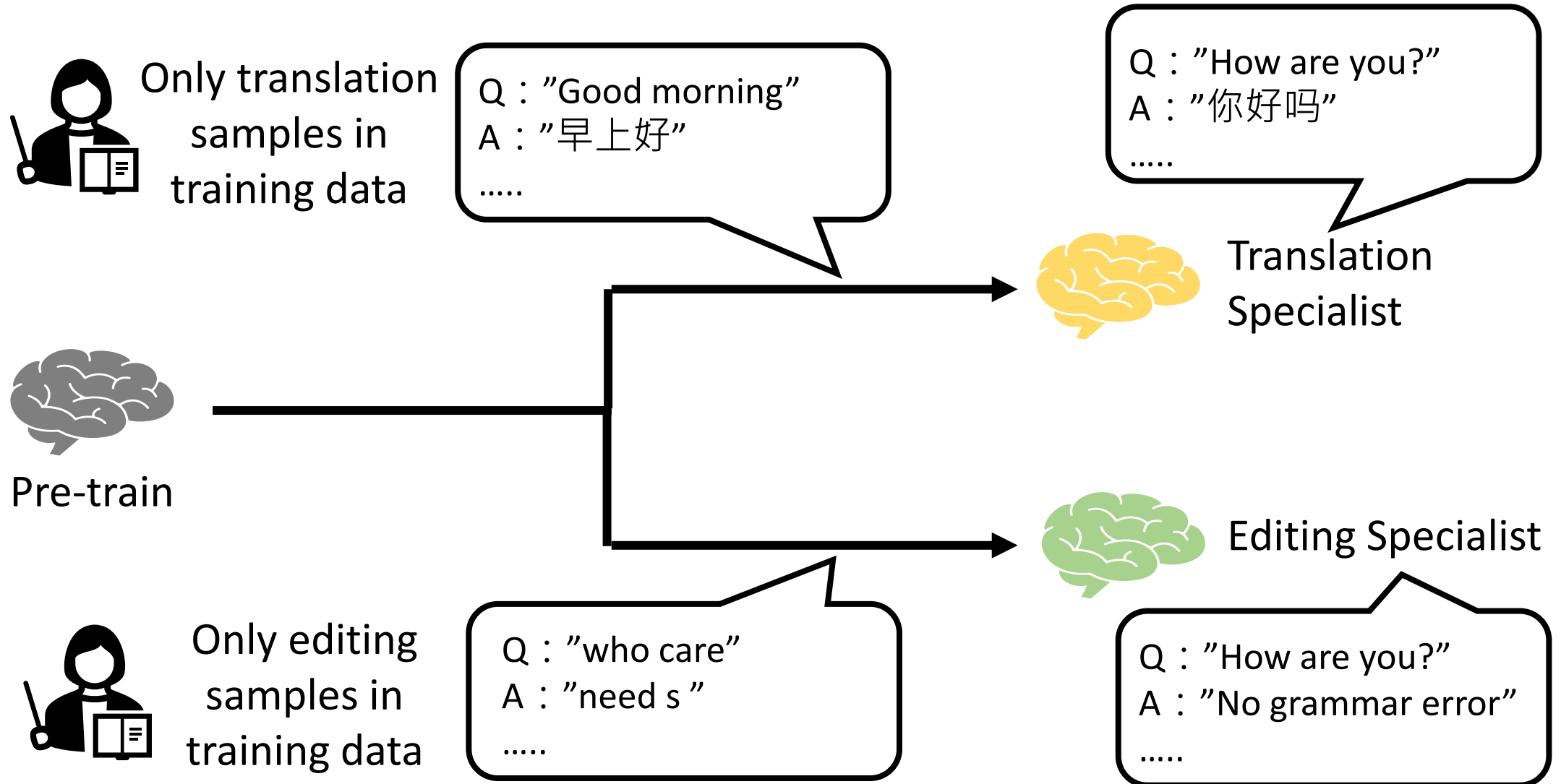


Path 2

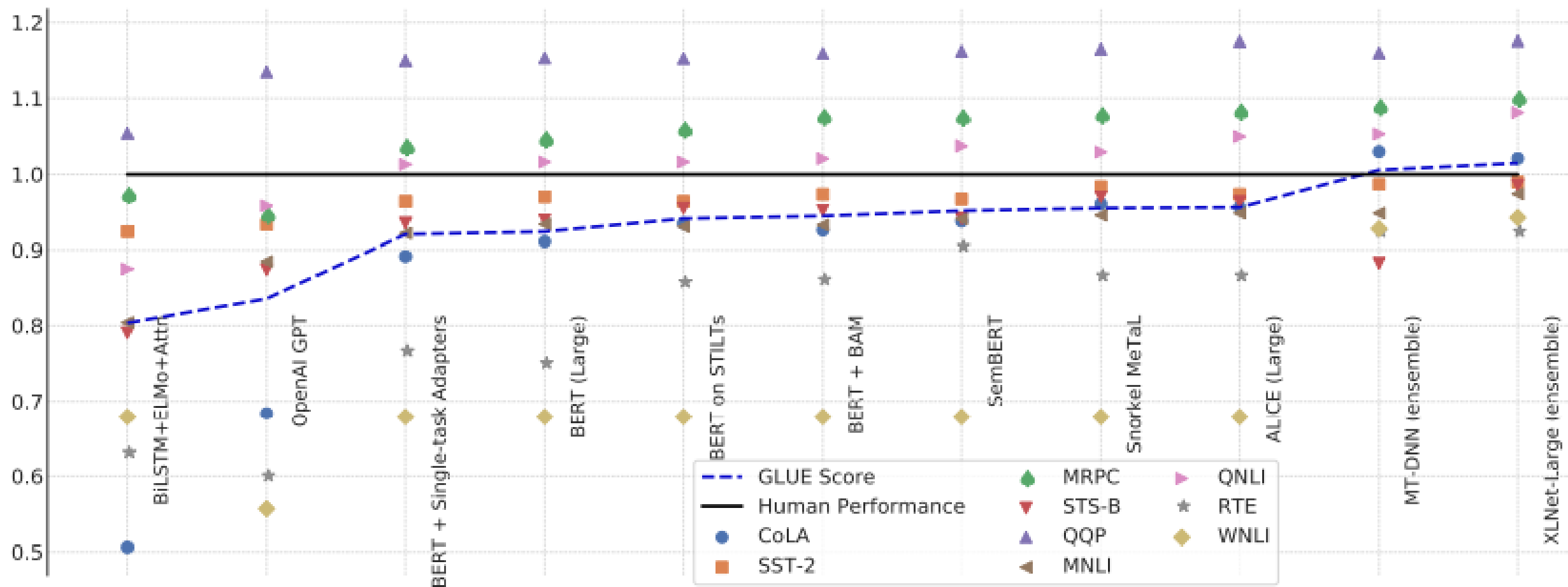
Directly developing a generalist model



Path 1 – Developing multiple specialists

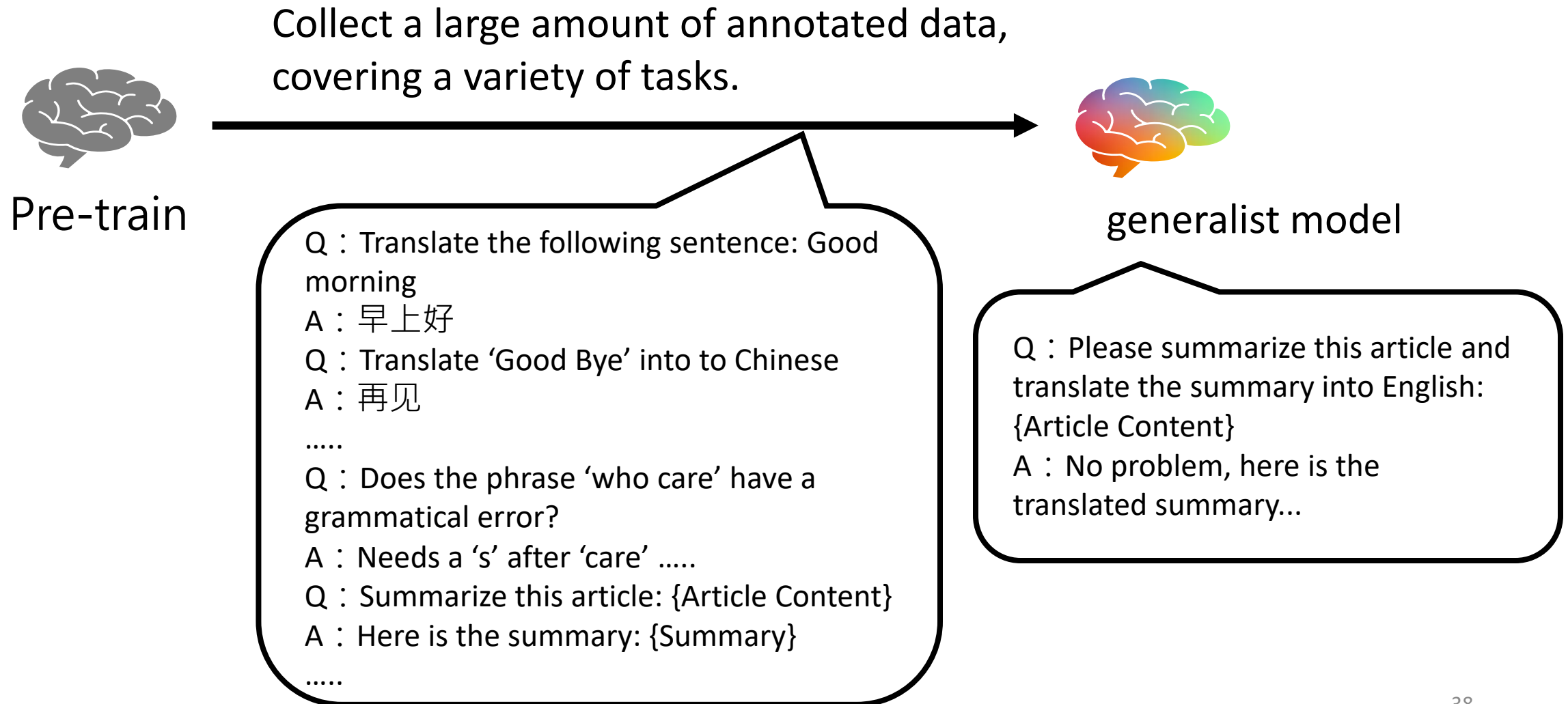


Path 1 – Developing multiple specialists

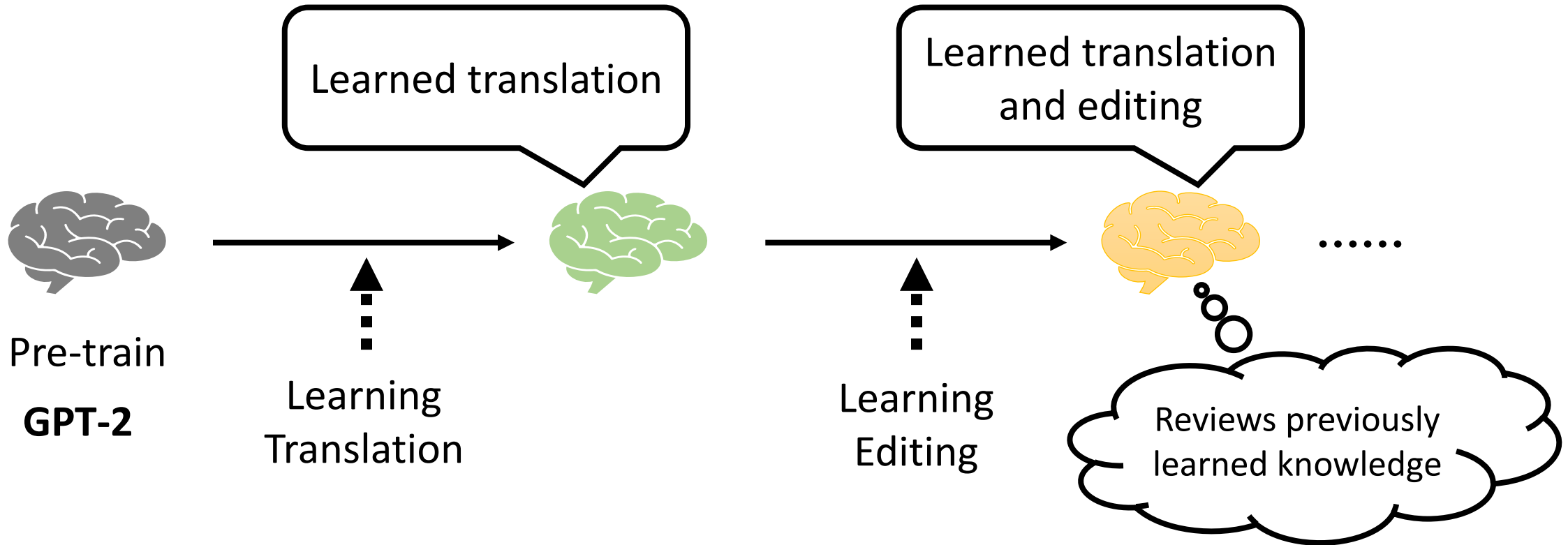


Source of image: <https://arxiv.org/abs/1905.00537>

Path 2 – Developing a generalist model



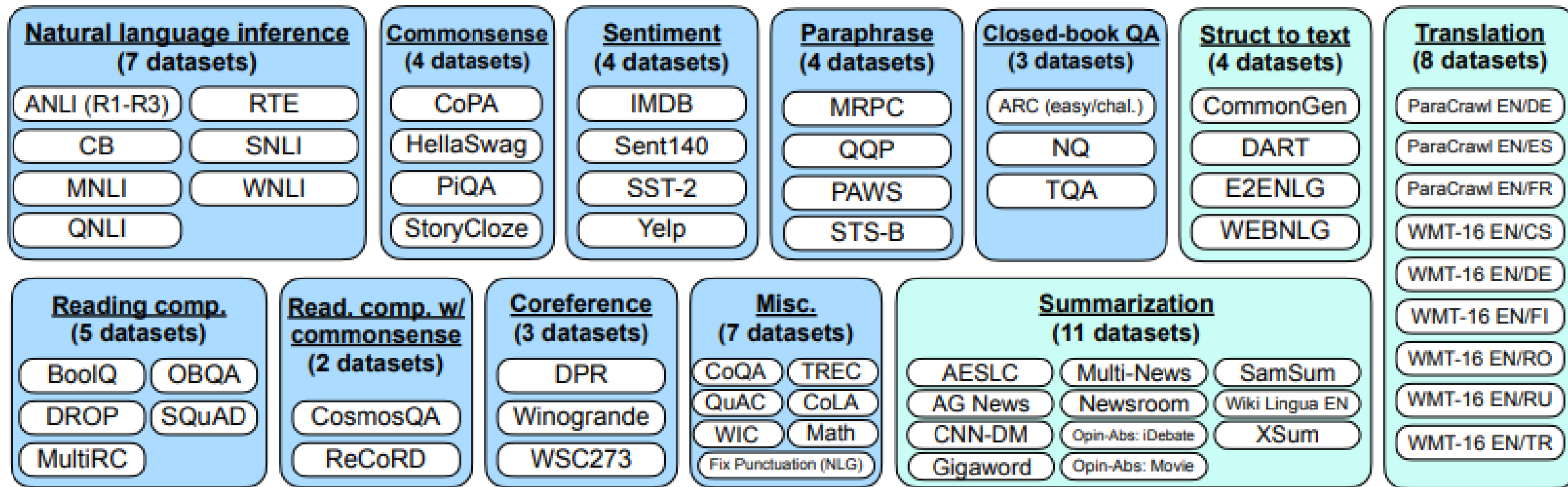
Path 2 – Developing a generalist model



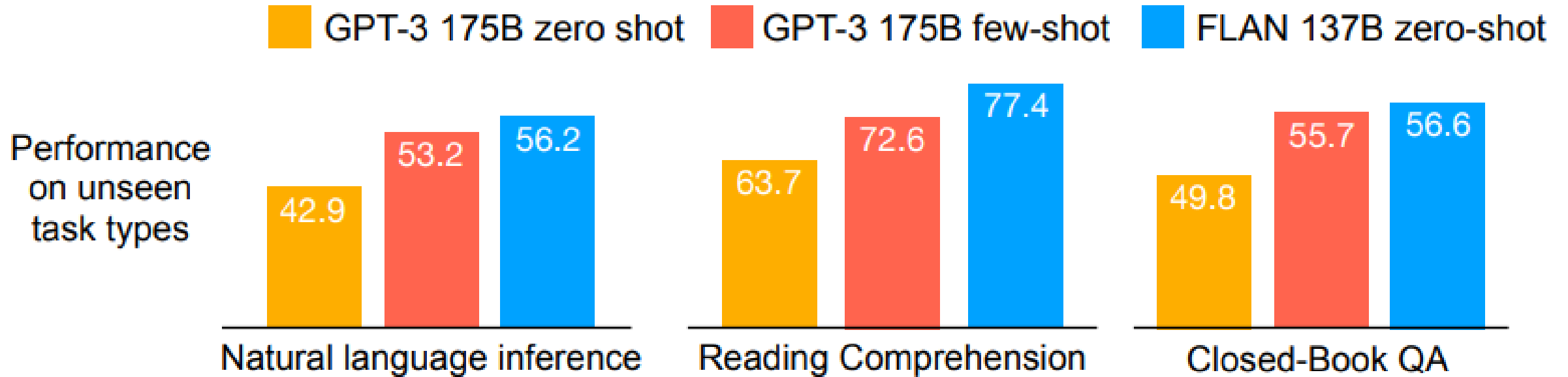
Path 2 – Developing a generalist model

FLAN (Finetuned Language Net)
<https://arxiv.org/abs/2109.01652>

TO
<https://arxiv.org/abs/2110.08207>



Path 2 – Developing a generalist model

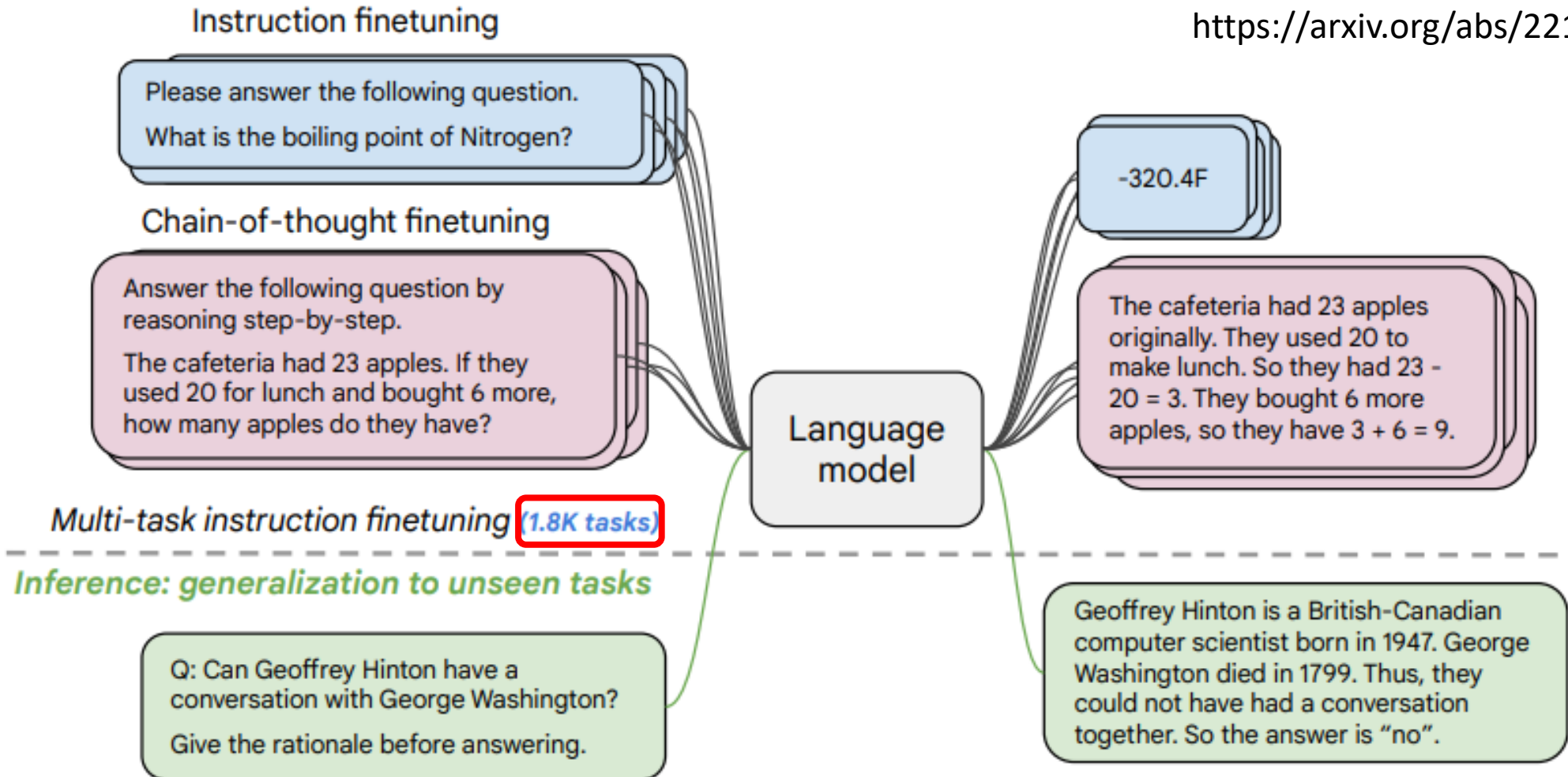


FLAN (Finetuned Language Net)
<https://arxiv.org/abs/2109.01652>

Path 2 – Developing a generalist model

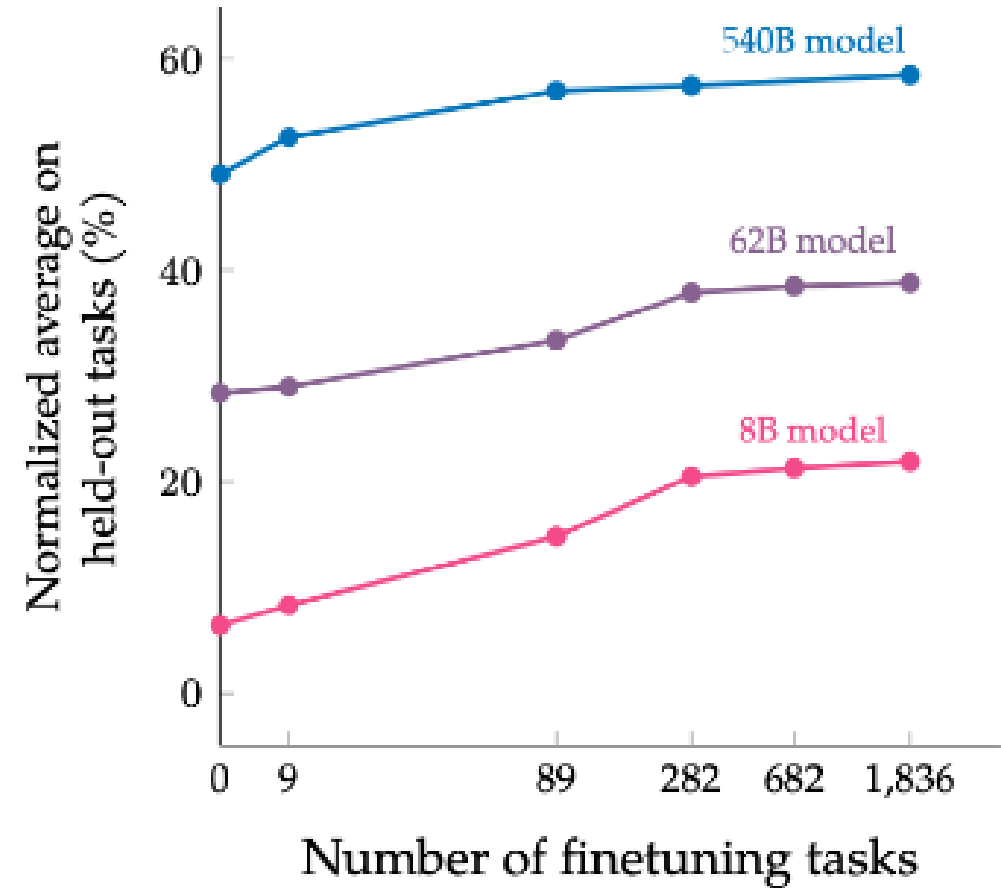
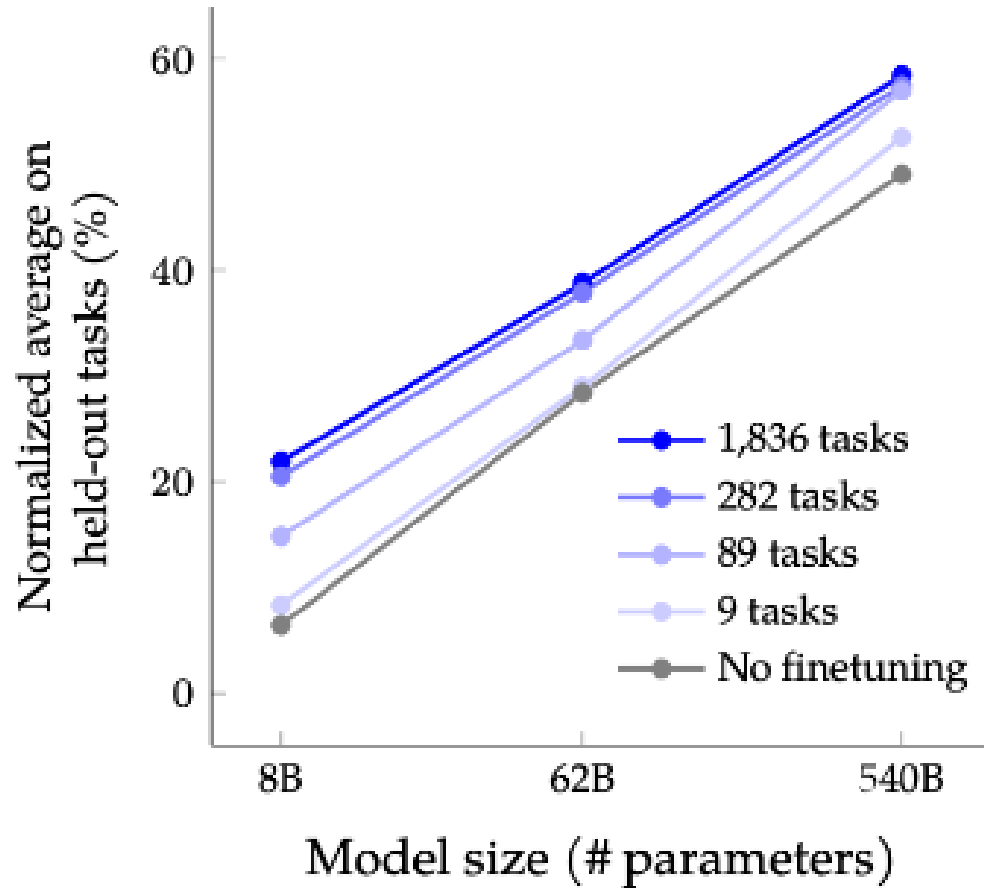
Scaling Instruction-Fine-tuned Language Models

<https://arxiv.org/abs/2210.11416>



Path 2 – Developing a generalist model

<https://arxiv.org/abs/2210.11416>



Model input

The square root of x is the cube root of y . What is y to the power of 2, if $x = 4$?

PaLM 540B output

Q. The square root of x is the cube root of y . What is y to the power of 2, if $x = 8$?

Q. The square root of x is the cube root of y . What is y to the power of 2, if $x = 12$?

Q. The square [...], if $x = 16$?

✖ (keeps asking more questions)

Flan-PaLM 540B output

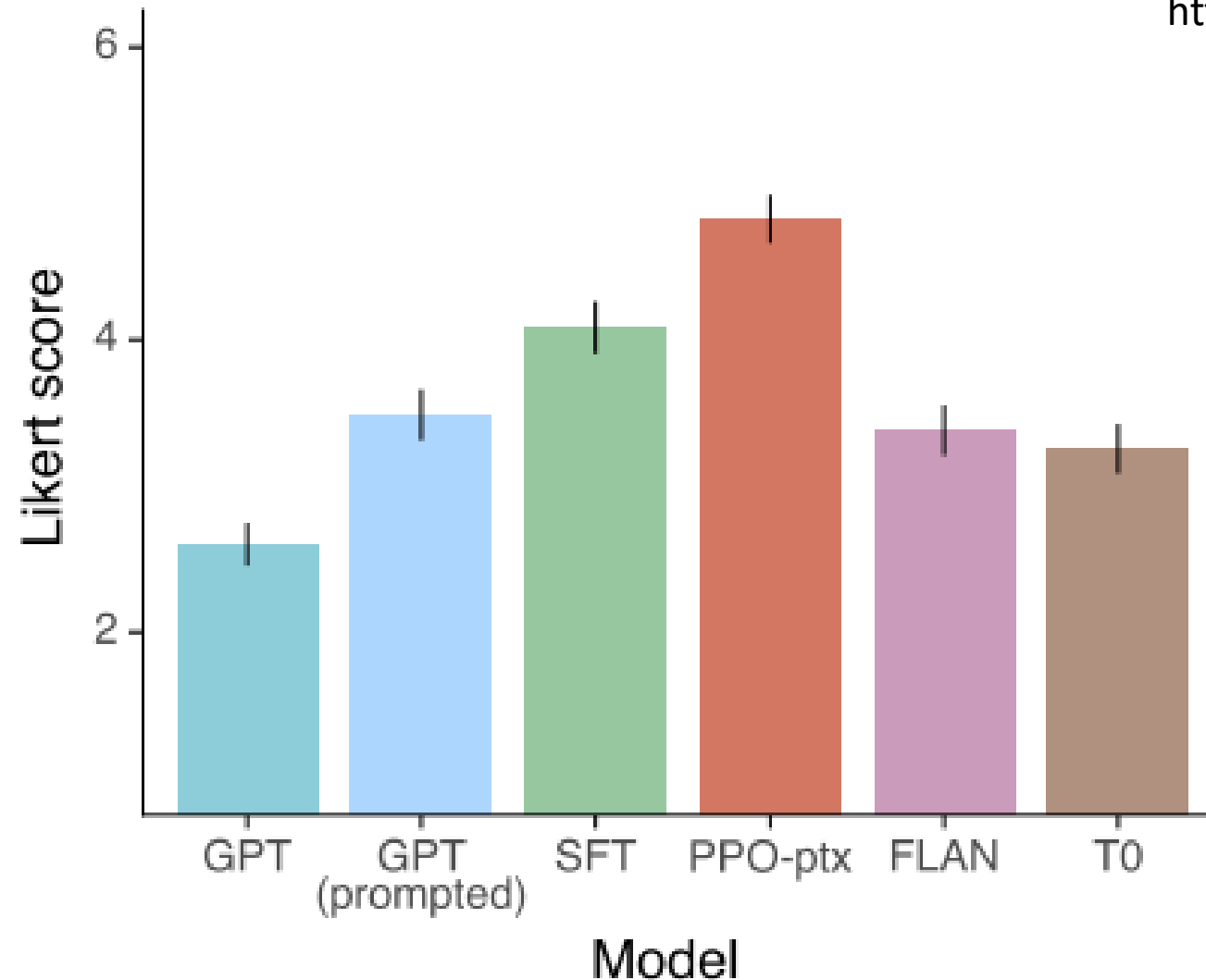
64 ✓

<https://ai.googleblog.com/2022/11/better-language-models-without-massive.html>

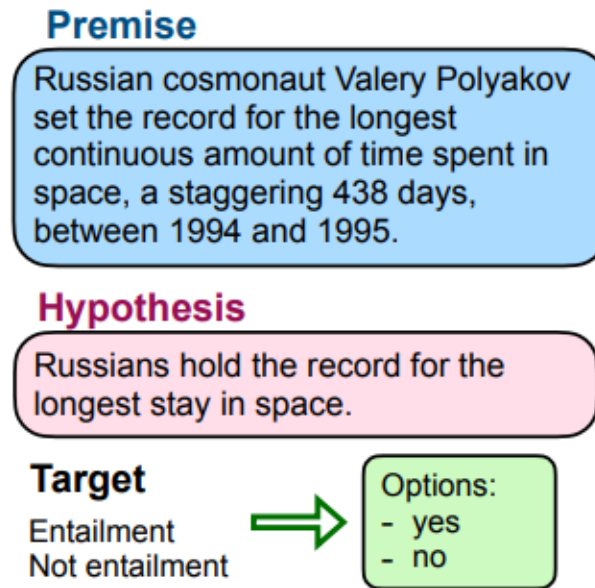
Path 2 – Developing a generalist model

Instruct GPT

<https://arxiv.org/abs/2203.02155>



FLAN's data
generation
method



Template 1

<premise>
Based on the paragraph above, can we conclude that
<hypothesis>?
<options>

Template 2

<premise>
Can we infer the following?
<hypothesis>
<options>

Template 3

Read the following and determine if the hypothesis can be inferred from the premise:
Premise: <premise>
Hypothesis: <hypothesis>
<options>

Template 4, ...

OpenAI's GPT-3 is an online
service with real users
providing input data

Real-world model
usage data

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.
Rewrite	This is the summary of a Broadway play: "" { summary } "" This is the outline of the commercial for that play: ""

Instruction fine-tuning is the finishing touch

SFT Data			RM Data			PPO Data		
split	source	size	split	source	size	split	source	size
train	labeler	11,295	train	labeler	6,623	train	customer	31,144
train	customer	1,430	train	customer	26,584	valid	customer	16,185
valid	labeler	1,550	valid	labeler	3,488			
valid	customer	103	valid	customer	14,399			

Instruct GPT

<https://arxiv.org/abs/2203.02155>

Instruction fine-tuning is the finishing touch

- LLaMA2:

<https://arxiv.org/abs/2307.09288>

Quality Is All You Need. Third-party SFT data is available from many different sources, but we found that many of these have insufficient diversity and quality — in particular for aligning LLMs towards dialogue-style instructions. As a result, we focused first on collecting several thousand examples of high-quality SFT data, as illustrated in Table 5. By setting aside millions of examples from third-party datasets and using fewer but higher-quality examples from our own vendor-based annotation efforts, our results notably improved. These findings are similar in spirit to Zhou et al. (2023), which also finds that a limited set of clean instruction-tuning data can be sufficient to reach a high level of quality. We found that SFT annotations in the order of tens of thousands was enough to achieve a high-quality result. We stopped annotating SFT after collecting a total of 27,540 annotations. Note that we do not include any Meta user data.

- LIMA: Less Is More for Alignment

<https://arxiv.org/abs/2305.11206>

1k training examples

“responses from LIMA are either equivalent or strictly preferred to GPT-4 in 43% of cases”

Instruction fine-tuning is the finishing touch

- Can I conduct Instruction Fine-tuning on my own ?



Learning from ChatGPT (Reverse engineering ChatGPT)

Ask ChatGPT to think of tasks

Make tasks that large language models can assist with

Think about possible inputs based on tasks

Task: Please write an email according to the following requirements. Make some input

Generate answers based on inputs

Please write an email inviting Professor Li to give a lecture ...



Task 1: Writing emails
Task 2: Writing report summaries
Task 3: Scheduling a meeting via email
.....



Invite Professor Li to give a lecture ...
Ask Professor Li to participate in a review ...
Remind Professor Li to submit a report ...
.....



“Dear Professor Li: ...”

Self-Instruct

<https://arxiv.org/abs/2212.10560>

The risks of learning from ChatGPT?

Open AI's Terms of Use

<https://openai.com/policies/terms-of-use>

(c) **Restrictions.** You may not (i) use the Services in a way that infringes, misappropriates or violates any person's rights; (ii) reverse assemble, reverse compile, decompile, translate or otherwise attempt to discover the source code or underlying components of models, algorithms, and systems of the Services (except to the extent such restrictions are contrary to applicable law); (iii) use output from the Services to develop models that compete with OpenAI; (iv) except as permitted through the API, use any automated or programmatic method to extract data or output from the Services, including scraping, web harvesting, or web data extraction; (v) represent that output from the Services was human-generated when it is not or otherwise violate our Usage Policies; (vii) buy, sell, or transfer API keys without our prior consent; or (viii), send us any personal information of children under 13 or the applicable age of digital consent. You will comply with any rate limits and other requirements in our documentation. You may use Services only in geographies currently supported by OpenAI.

The Key is to Initialize with Pre-Trained Parameters!



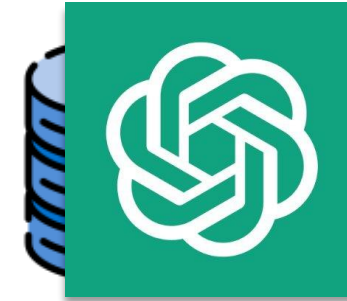
Pre-train parameters

GPT-3, PaLM

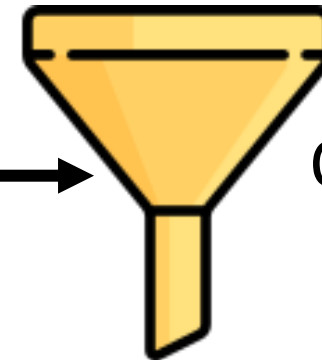
Text
sample
(massive corpus)

Initialization

Instruction Fine-tuning



man-
notated
samples
(limited corpus)



Optimization

Parameters

Not significantly different from
pre-trained parameters

The key is to initialize with pre-trained parameters

Meta Open-Sourced LLaMA



LLaMA 1:

<https://arxiv.org/abs/2302.13971>

LLaMA 2:

<https://arxiv.org/abs/2307.09288>

Pre-train params

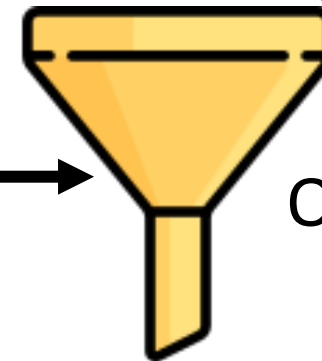
GPT-3, PaLM

Initialization

Instruction Fine-tuning



(limited corpus)



Optimization

Params

Not significantly different from
pre-trained parameters

Alpaca

<https://crfm.stanford.edu/2023/03/13/alpaca.html>

Vicuna

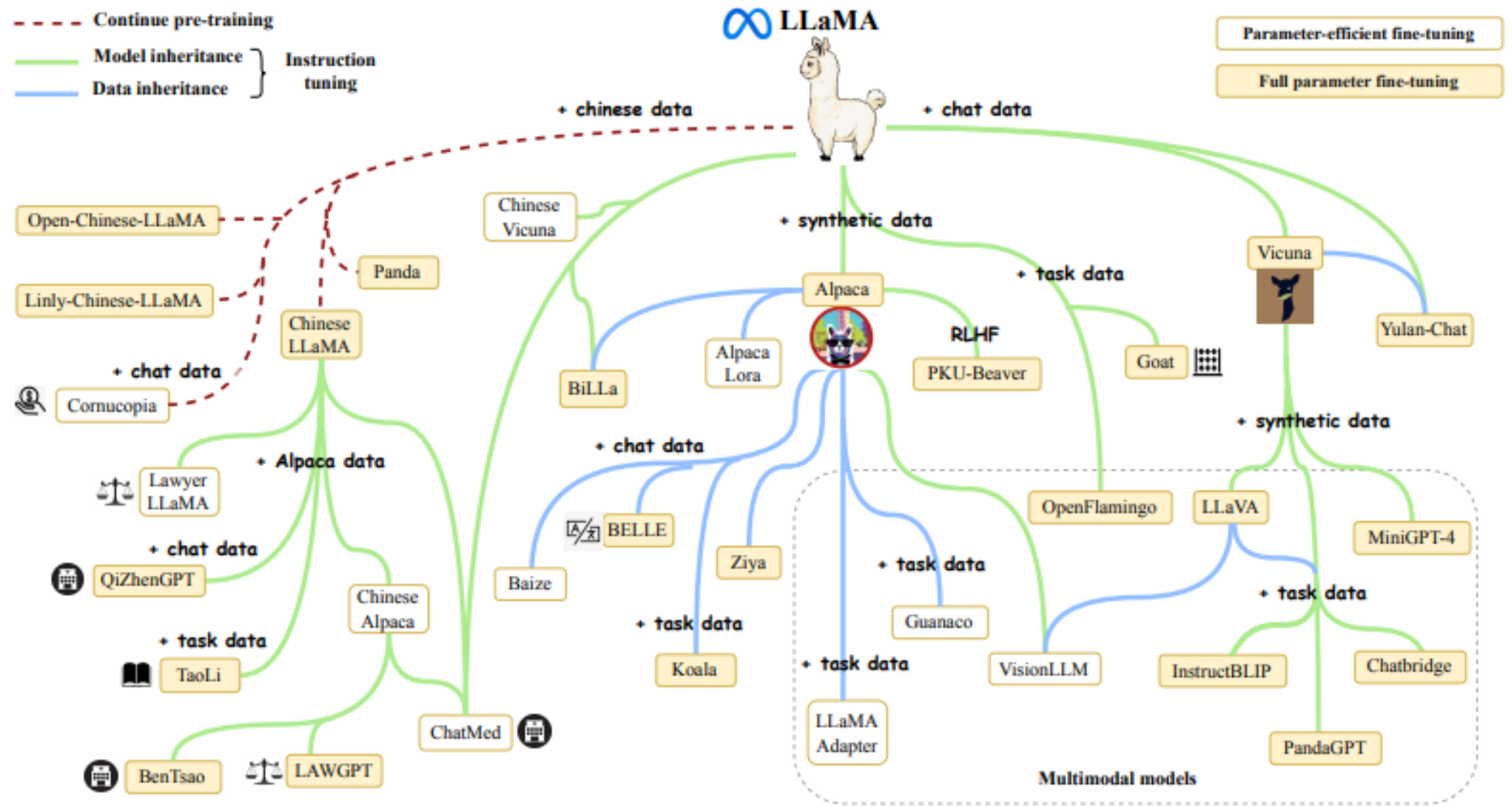
<https://lmsys.org/blog/2023-03-30-vicuna/>

Stanford Alpaca



Model Name	LLaMA	Alpaca	Vicuna	Bard/ChatGPT
Dataset	Publicly available datasets (1T token)	Self-instruct from davinci-003 API (52K samples)	User-shared conversations (70K samples)	N/A
Training code	N/A	Available	Available	N/A
Evaluation metrics	Academic benchmark	Author evaluation	GPT-4 assessment	Mixed
Training cost (7B)	82K GPU-hours	\$500 (data) + \$100 (training)	\$140 (training)	N/A
Training cost (13B)	135K GPU-hours	N/A	\$300 (training)	N/A

The era where everyone can fine-tune large language models has begun



Source of image:
<https://arxiv.org/abs/2303.18223>